Self similar diffusion

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1 The self-similar framework

1.1 Rescaling into a self-similar frame

Let $u_i^*(\mathbf{x}^*, t^*)$, $i = 1, \ldots, d$, satisfy an evolution equation of the form

$$\frac{\partial u_i^*}{\partial t^*} = R_i^*(u^*), \tag{1}$$

on $\mathbb{R}^d$. We rescale amplitude, length, and time by

$$u_i = \frac{u_i^*}{U}, \qquad x_j = \frac{x_j^*}{L}, \qquad t = \frac{t^*}{s(t)}, \tag{2}$$

where the amplitude scale U and the length scale L depend on the non dimensional time t. The timescale is determined by specifiying the clock

$$\tau = \frac{\mathrm{d}t^*}{\mathrm{d}t}. \tag{3}$$

The clock τ is problem specific and can contain the time dependent length and amplitude scales. It is defined in §4.1 and §5.1 for the two cases considered here.

Throughout, a prime denotes differentiation with respect to t and we use the following abbreviations

$$\sigma_L = \frac{L'}{L}, \qquad \sigma_U = \frac{U'}{U}. \tag{4}$$

Differentiating $x_j = x_j^*/L$ with respect to t at fixed x_j^* gives, for each component j,

$$\frac{\mathrm{d}x_j}{\mathrm{d}t} = -x_j^* \frac{L'}{L^2} = -x_j\, \sigma_L. \tag{5}$$

Applying the chain rule to the left-hand side of (1), and using (3) and (5),

$$\begin{aligned} \frac{\partial u_i^*}{\partial t^*} &= \frac{\mathrm{d}t}{\mathrm{d}t^*} \left(\frac{\partial (U u_i)}{\partial t} + U \frac{\partial u_i}{\partial x_j} \frac{\mathrm{d}x_j}{\mathrm{d}t} \right) \\ &= \frac{U}{\tau} \left(\frac{\partial u_i}{\partial t} + \sigma_U\, u_i - \sigma_L\, x_j \frac{\partial u_i}{\partial x_j} \right). \end{aligned} \tag{6}$$

Equating (6) to R_i^* and solving for $\partial u_i/\partial t$ gives the result used throughout this document. Under the rescaling (2), the equation (1) becomes

$$\frac{\partial u_i}{\partial t} = \underbrace{\sigma_L\, x_j \frac{\partial u_i}{\partial x_j} - \sigma_U\, u_i}_{\text{scaling terms}} + \underbrace{\frac{\tau}{U}\, R_i^*(Uu,\, Lx)}_{\text{physical terms}}. \tag{7}$$

Accordingly we split (7) into two flows. The *scaling operator* $\Phi_{\Delta t}^{(\mathrm{S})}$ advances the scaling terms,

$$\Phi_{\Delta t}^{(\mathrm{S})}(u_i) = u_i + \int_t^{t+\Delta t} \left(\sigma_L\, x_j \frac{\partial u_i}{\partial x_j} - \sigma_U\, u_i \right) \mathrm{d}t, \tag{8}$$

and the *physics operator* $\Phi^{(\mathrm{P})}_{\Delta t}$ advances the rest,

$$\Phi^{(\mathrm{P})}_{\Delta t}(u_i) = u_i + \int_t^{t+\Delta t} \frac{\tau}{U}\, R_i^*(Uu,\, Lx)\, \mathrm{d}t. \tag{9}$$

Both are written as the field plus the integral of the terms they advance. $\Phi^{(\mathrm{S})}$ is obtained in closed form below and is shared; $\Phi^{(\mathrm{P})}$ is equation-specific and is supplied in §4.2 and §5.6.

1.2 The scaling operator

Evaluating $\Phi^{(\mathrm{S})}_{\Delta t}$ amounts to solving

$$\frac{\partial u_i}{\partial t} = \sigma_L\, x_j \frac{\partial u_i}{\partial x_j} - \sigma_U\, u_i. \tag{10}$$

This is a linear, first-order PDE and can be solved by the method of characteristics. Note that the scaling terms do not couple the components of u_i, so the argument below is componentwise.

Let $x_j(t)$ denote a characteristic curve satisfying

$$\frac{\mathrm{d}x_j}{\mathrm{d}t} = -x_j\, \sigma_L. \tag{11}$$

The total derivative of u_i along such a curve is

$$\frac{\mathrm{d}u_i}{\mathrm{d}t} = \frac{\partial u_i}{\partial t} + \frac{\partial u_i}{\partial x_j}\frac{\mathrm{d}x_j}{\mathrm{d}t} = \frac{\partial u_i}{\partial t} - \sigma_L\, x_j \frac{\partial u_i}{\partial x_j}, \tag{12}$$

so along the characteristic (10) reduces to the ordinary differential equation

$$\frac{\mathrm{d}u_i}{\mathrm{d}t} = -\sigma_U\, u_i. \tag{13}$$

Integrating (11) from t_k to t_{k+1} gives

$$\ln x_j(t_{k+1}) - \ln x_j(t_k) = -\big[\ln L_{k+1} - \ln L_k\big] \quad \Longrightarrow \quad x_j(t_{k+1}) = x_j(t_k)\, \frac{L_k}{L_{k+1}}, \tag{14}$$

so that the characteristic arriving at the evaluation point $x_j \equiv x_j(t_{k+1})$ originated at

$$x_j(t_k) = x_j\, \frac{L_{k+1}}{L_k}. \tag{15}$$

Integrating (13) gives

$$u_i(t_{k+1}) = u_i(t_k)\, U_k/U_{k+1}. \tag{16}$$

Combining this with the characteristic origin (15) and the initial condition $u_i(t_k, \cdot)$ gives the scaling operator in closed form,

$$\Phi^{(\mathrm{S})}_{\Delta t}(u_i)(\mathbf{x}) = \frac{U_k}{U_{k+1}}\, u_i\!\left(\mathbf{x}\, \frac{L_{k+1}}{L_k}\right). \tag{17}$$

Writing the step ratios

$$\lambda_{U,k+1} = \frac{U_{k+1}}{U_k}, \qquad \lambda_{L,k+1} = \frac{L_{k+1}}{L_k}, \tag{18}$$

this is

$$\Phi^{(\mathrm{S})}_{\Delta t}(u_i)(\mathbf{x}) = \frac{1}{\lambda_{U,k+1}}\, u_i(\mathbf{x}\, \lambda_{L,k+1}). \tag{19}$$

1.2.1 The scaling operator in $\boldsymbol{\kappa}$-space

For equations formulated spectrally it is useful to have the transform of the scaling terms. Using $\widehat{\cdot}$ for the spatial Fourier transform,

$$\hat{u}_i(\boldsymbol{\kappa}) = \int u_i(\mathbf{x})\, e^{-\mathrm{i}\kappa_j x_j}\, \mathrm{d}\mathbf{x}, \tag{20}$$

and writing the spread term in conservative form,

$$x_j \frac{\partial u_i}{\partial x_j} = \frac{\partial}{\partial x_j}\left(x_j u_i\right) - d\, u_i, \tag{21}$$

gives

$$\widehat{x_j \frac{\partial u_i}{\partial x_j}} = \mathrm{i}\kappa_j \left(\mathrm{i}\frac{\partial \hat{u}_i}{\partial \kappa_j}\right) - d\,\hat{u}_i = -\left(\kappa_j \frac{\partial \hat{u}_i}{\partial \kappa_j} + d\,\hat{u}_i\right), \tag{22}$$

so that the transform of (10) is

$$\frac{\partial \hat{u}_i}{\partial t} = -\sigma_L \left(\kappa_j \frac{\partial \hat{u}_i}{\partial \kappa_j} + d\,\hat{u}_i\right) - \sigma_U\, \hat{u}_i. \tag{23}$$

This is again a linear, first-order PDE, now with characteristics in $\boldsymbol{\kappa}$ satisfying

$$\frac{\mathrm{d}\kappa_j}{\mathrm{d}t} = \kappa_j\, \sigma_L, \tag{24}$$

along which (23) reduces to

$$\frac{\mathrm{d}\hat{u}_i}{\mathrm{d}t} = -\left(d\,\sigma_L + \sigma_U\right) \hat{u}_i. \tag{25}$$

Integrating (24) from t_k to t_{k+1} gives $\kappa_j(t_{k+1}) = \kappa_j(t_k)\,\lambda_{L,k+1}$, so the characteristic arriving at $\boldsymbol{\kappa}$ originated at $\boldsymbol{\kappa}/\lambda_{L,k+1}$, while integrating (25) gives $\hat{u}_i(t_{k+1}) = \hat{u}_i(t_k)/(\lambda_{U,k+1}\lambda_{L,k+1}^{d})$. The scaling operator in $\boldsymbol{\kappa}$-space is therefore

$$\widehat{\Phi^{(\mathrm{S})}_{\Delta t}(u_i)}(\boldsymbol{\kappa}) = \frac{1}{\lambda_{U,k+1}\,\lambda_{L,k+1}^{d}}\, \hat{u}_i\!\left(\frac{\boldsymbol{\kappa}}{\lambda_{L,k+1}}\right). \tag{26}$$

This agrees with transforming (19) directly: the substitution $\mathbf{x}' = \mathbf{x}\,\lambda_{L,k+1}$ in (20) contributes the Jacobian $\lambda_{L,k+1}^{-d}$ and maps $\boldsymbol{\kappa}$ to $\boldsymbol{\kappa}/\lambda_{L,k+1}$. A growing length scale ($\lambda_{L,k+1} > 1$) thus compresses u_i toward $\mathbf{x} = 0$ and correspondingly stretches $\hat{u}_i$ toward larger $|\boldsymbol{\kappa}|$.

1.3 Spectral interpolation

The closed form (19) requires u_i at the rescaled points $\mathbf{x}\,\lambda_{L,k+1}$, which are not grid points. On a uniform periodic grid the represented field is band-limited, so it can be evaluated at arbitrary points exactly, with no interpolation error, by summing its truncated Fourier series there:

$$u_i(\mathbf{x}\,\lambda_{L,k+1}) = \sum_{\boldsymbol{\kappa}\in\mathcal{K}} \hat{u}_i(\boldsymbol{\kappa})\, \exp(\mathrm{i}\,\kappa_j x_j\, \lambda_{L,k+1}), \tag{27}$$

where $\mathcal{K}$ is the set of represented wavenumbers and $\hat{u}_i(\boldsymbol{\kappa})$ are the normalized transform coefficients of u_i. This is the defining property of the scheme: the scaling step introduces no spatial discretization error of its own.

Queries outside the box. The true problem we are trying to approximate is posed on an infinite box, with u_i going to zero as it approaches the boundaries. For $\lambda_{L,k+1} > 1$ it is possible that $|x_j\,\lambda_{L,k+1}| > H/2$, in which case we would like the rescaling in (27) to return 0. However, this is not the current behavior: instead, it wraps around the periodic box and returns a potentially nonzero value from the other end of the domain. Therefore, to obtain the desired behavior we modify the scaling step to

$$u_i(\mathbf{x}\,\lambda_{L,k+1}) = \chi(\mathbf{x}\,\lambda_{L,k+1}) \sum_{\boldsymbol{\kappa}\in\mathcal{K}} \hat{u}_i(\boldsymbol{\kappa})\, \exp(\mathrm{i}\,\kappa_j x_j\, \lambda_{L,k+1}), \qquad \chi(\mathbf{y}) = \prod_d \mathbb{1}[\,|y_d| \le H/2\,]. \tag{28}$$

1.4 Constraint functionals

We will set the ratios $\lambda_{U,k+1}, \lambda_{L,k+1}$ that the scaling step (19) depends on by requiring the rescaled field to carry prescribed values of two functionals F_1, F_2 of the form

$$F_m\left[\frac{1}{\lambda_U}\,u(x\lambda_L)\right] = \lambda_U^{-p_m}\,\lambda_L^{-q_m}\,F_m[u], \qquad m = 1, 2. \tag{29}$$

The targets F_1, F_2 are fixed once, from the initial condition (§2.2). Given a field u with current values $F_m^{\text{in}} = F_m[u]$, imposing $\lambda_U^{-p_m}\lambda_L^{-q_m}F_m^{\text{in}} = F_m$ and taking logarithms gives the linear system

$$\begin{bmatrix} p_1 & q_1 \\ p_2 & q_2 \end{bmatrix} \begin{bmatrix} \ln\lambda_U \\ \ln\lambda_L \end{bmatrix} = \begin{bmatrix} \ln F_1^{\text{in}} - \ln F_1 \\ \ln F_2^{\text{in}} - \ln F_2 \end{bmatrix}, \tag{30}$$

which is uniquely solvable when $p_1q_2 - p_2q_1 \neq 0$, i.e. when each functional responds to amplitude and to length rescaling differently from the other, and is solved in closed form. Once F_m^{in} is evaluated, no further evaluation of the scaled field is needed to find the ratios.

In both cases the second functional is a second moment, and we normalize it by the target value of the zeroth moment F_1. As the normalization is a constant, it changes neither the exponents nor the ratios the constraints produce; it gives F_2 the units of a squared length, a mean-square width. The functionals themselves are equation-specific (§4.3, §5.8); the substep that applies (30) is ConstrainedScale (§2.5).

1.5 Power-law ansatz and the step in physical time

The algorithm also needs to *predict* scale ratios over a fraction of a step, at points where no constraint is available to fix them (§2.7). For this we assume that both scales follow a power law in physical time,

$$X = A_X\,(t^* + \alpha_X)^{\beta_X}, \qquad X \in \{U, L\}, \tag{31}$$

with a prefactor $A_X > 0$, an offset α_X, and an exponent β_X. The prefactor cannot be dropped: the scales are normalized to $U_0 = L_0 = 1$ at the initial condition (§2.2), before the flow has reached the self-similar state, so nothing ties that normalization to the one of the power law. The prefactor cancels from the ratio over a fraction i of step k,

$$\frac{X_{k+i}}{X_k} = \left(1 + \frac{i\,\Delta t_k^*}{t_k^* + \alpha_X}\right)^{\beta_X}. \tag{32}$$

Since the time step $\Delta t_k = t_{k+1} - t_k$ is taken in the self-similar time t, the corresponding physical increment must be obtained from the clock,

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \tau\,\mathrm{d}t, \tag{33}$$

which can be approximated with a three-point Adams–Bashforth-style quadrature:

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \tau\,\mathrm{d}t \approx w_0\,\tau_k + w_1\,\tau_{k-1} + w_2\,\tau_{k-2}. \tag{34}$$

The coefficients can be computed as follows.

$$\begin{aligned} w_0 &= \frac{h^3/3 + (2h_1 + h_2)\,h^2/2 + h_1(h_1 + h_2)\,h}{h_1(h_1 + h_2)}, \\ w_1 &= -\frac{h^3/3 + (h_1 + h_2)\,h^2/2}{h_1 h_2}, \\ w_2 &= \frac{h^3/3 + h_1\,h^2/2}{h_2(h_1 + h_2)}, \end{aligned} \tag{35}$$

where $h = \Delta t_k$, $h_1 = \Delta t_{k-1}$, $h_2 = \Delta t_{k-2}$. In this work the step is constant, $\Delta t_k = \Delta t$ for every k (§2.3). With $h = h_1 = h_2 = \Delta t$ the weights reduce to those of the third-order Adams–Bashforth method,

$$w_0 = \frac{23}{12}\,\Delta t, \qquad w_1 = -\frac{16}{12}\,\Delta t, \qquad w_2 = \frac{5}{12}\,\Delta t. \tag{36}$$

Eliminating $t_k^* + \alpha_X = (X_k/A_X)^{1/\beta_X}$ from (32) and taking logarithms gives

$$\ln \frac{X_{k+i}}{X_k} = \beta_X \ln\!\left(1 + \exp\!\left(\ln i + \ln \Delta t_k^* - \frac{\ln X_k - \ln A_X}{\beta_X}\right)\right). \tag{37}$$

The algorithm uses this with $\ln A_X = 0$, as the function

$$\ln \frac{X_{k+i}}{X_k} = G_X(i, \beta_X; k), \qquad G_X(i, \beta; k) = \beta \ln\!\left(1 + \exp\!\left(\ln i + \ln \Delta t_k^* - \frac{\ln X_k}{\beta}\right)\right), \tag{38}$$

in both directions: it is evaluated forward, with $i = \frac{1}{2}$, to predict a half-step ratio (§2.5), and inverted for β, with $i = 1$, to estimate the exponents from the realized history (§2.8).

The prefactor within a step. Setting $\ln A_X = 0$ is not an assumption that $A_X = 1$; the step does not need A_X. By (32) the ratio depends on the pair $(\beta_X,\, t_k^* + \alpha_X)$, and expanding in the step,

$$\ln \frac{X_{k+i}}{X_k} = \frac{\beta_X}{t_k^* + \alpha_X}\, i\, \Delta t_k^* \left(1 - \frac{i\, \Delta t_k^*}{2\,(t_k^* + \alpha_X)} + \dots\right). \tag{39}$$

The leading term depends on the pair only through $\beta_X/(t_k^* + \alpha_X)$, which is the local rate σ_X/τ (§3.1). The inversion of §2.8 fixes exactly this combination, by matching the realized ratio of the step just taken, whatever $\ln A_X$ is assumed. With $\ln A_X = 0$ the β that does so is an *effective* exponent $\tilde\beta_X$ that absorbs the prefactor: matching $(\ln X_k - \ln A_X)/\beta_X$ with $\ln X_k/\tilde\beta_X$ gives $\tilde\beta_X \approx \beta_X\,(1 + \ln A_X/\ln X_k)$, which tends to β_X as the scales grow. The prefactor only decides how the fixed $\beta_X/(t_k^* + \alpha_X)$ is split between β_X and $t_k^* + \alpha_X$, which enters the correction in (39); that is smaller than the leading term by $i\,\Delta t_k^*/(t_k^* + \alpha_X) = i\,\sigma_X \Delta t_k/\beta_X$, the order of the splitting error itself. The algorithm therefore carries only the effective exponents, and A_X, α_X, β_X are recovered separately, from the whole history, in §3.

Logarithmic arithmetic. U_k and L_k may become extremely large or small, so we store $\ln U_k$ and $\ln L_k$ rather than U_k and L_k, and perform all arithmetic on them in the logarithmic number system, where multiplication and division become addition and subtraction, and where sums and differences are evaluated by

$$\log_b(X+Y) = \log_b(X) + \log_b\!\big(1 + b^{\,y-x}\big), \qquad \log_b(X-Y) = \log_b(X) + \log_b\!\big(1 - b^{\,y-x}\big) \quad (X > Y > 0), \tag{40}$$

with $x = \log_b(X)$, $y = \log_b(Y)$. In particular (34) is evaluated as

$$\ln \Delta t_k^* = \ln\!\big(w_0\, e^{\ln \tau_k} + w_1\, e^{\ln \tau_{k-1}} + w_2\, e^{\ln \tau_{k-2}}\big). \tag{41}$$

2 Algorithm

The integrator below is stated once, for any equation that supplies the three items of §1: a clock τ, a physics operator $\Phi^{(\mathrm{P})}$, and a pair of constraint functionals. Everything else — the splitting, the scale bookkeeping, the exponent estimation — is common.

2.1 Domain, grid, and notation

The spatial domain is periodic with a fixed physical side length H and N grid points per direction, centred at the origin, so that $x_j \in [-H/2, H/2)$ with uniform spacing $\Delta x = H/N$ and grid points

$$x_n = -\frac{H}{2} + n\,\Delta x, \qquad n = 0, 1, \dots, N-1 \tag{42}$$

in each of the d directions. Every integral over the domain below is approximated by the Riemann sum

$$\int f(\mathbf{x})\,\mathrm{d}\mathbf{x} \approx (\Delta x)^d \sum_{\mathbf{n}} f(\mathbf{x_n}) \tag{43}$$

on this grid.

An index range abbreviates the corresponding tuple of stored values, e.g. $\ln \tau_{k-2:k} = (\ln \tau_{k-2}, \ln \tau_{k-1}, \ln \tau_k)$. The run constants H, N, the time step Δt of §2.3, the constraint targets F_1, F_2, and whatever physical parameters the equation carries are visible to every subroutine and are omitted from their signatures. The step is constant, $\Delta t_k = \Delta t$; the subscript is kept where a formula holds for a general step. As set out in §1.5, all arithmetic on the scales is done on $\ln U_k, \ln L_k$.

2.2 Initialization

A run is specified by the initial field u_0, the domain parameters H, N, the time step Δt of §2.3, a time horizon T_{end}, the splitting scheme *scheme* $\in \{\textsc{LieTrotter}, \textsc{Strang}\}$ (§2.6, §2.7), for equations that define one, the switch *project* of the optional projection inside Scale (§2.5), and initial guesses $\beta_U^{(0)}, \beta_L^{(0)}$, which only seed the bracket of the first root-find for β_U, β_L (§2.8). From these:

- The integration loop starts at $t_0 = 0$ and runs while $t_k < T_{\text{end}}$.
- The self-similar frame is initialized at the initial condition itself, $L_0 = U_0 = 1$, i.e. $\ln L_0 = \ln U_0 = 0$.
- The *target* values F_1, F_2 of the constraint functionals (§1.4) are fixed once, from u_0 itself,

$$F_1 = F_1[u_0], \qquad F_2 = F_2[u_0]. \tag{44}$$

- The flag β_{ok}, which records whether the current β_U, β_L are usable by Strang, starts as $\beta_{\text{ok}} = \text{false}$.

2.3 Time step

Every step is taken at the same step $\Delta t_k = \Delta t$, a run parameter, so that $t_k = k\,\Delta t$ and the quadrature (34) uses the constant weights (36). A constant step also keeps the samples of every recorded quantity equally spaced in t, which simplifies the time averages of Appendix C.

Nothing adapts the step to the field. When the physics operator has a state-dependent stability limit, Δt must be chosen small enough for the field the run settles into, and whether it is is monitored rather than enforced. Let $\Delta t_{\text{CFL}}(u)$ be the largest step at which Evolve from u stays within the stability limit of its time integrator; for Navier–Stokes it is the advective bound (194). The *realized* CFL number of step k,

$$\text{CFL}_k = \frac{\Delta t}{\Delta t_{\text{CFL}}(u_{k+1})}, \tag{45}$$

measures the step against the field it produced. It is recorded every step, and the run warns the first time it exceeds 1; the remedy is a smaller Δt. The 1D diffusion equation has no such limit, since its propagator (71) is exact at any step.

The end of the run. The loop runs while $t_k < T_{\text{end}}$, and the last step is *not* shortened to land on T_{end}, which would break the constant step. The run overshoots T_{end} by less than one step, which is immaterial on the logarithmic clock t. For the same reason, output taken at prescribed times t is taken at the first step reaching each time, never by shortening a step.

2.4 Top-level integrator

Algorithm 1 Top-level integrator

Require: u_0, H, N, Δt, T_{end}, *scheme*, $\beta_U^{(0)}$, $\beta_L^{(0)}$

$t_0 \leftarrow 0$, $k \leftarrow 0$
$\ln U_0 \leftarrow 0$, $\ln L_0 \leftarrow 0$
Compute constraint targets F_1, F_2 from u_0 via (44)
$\beta_U \leftarrow \beta_U^{(0)}$, $\beta_L \leftarrow \beta_L^{(0)}$, $\beta_{\text{ok}} \leftarrow$ false
Store $(u_0, \ln U_0, \ln L_0, \ln \tau_0, \beta_U, \beta_L, \beta_{\text{ok}})$
while $t_k < T_{\text{end}}$ **do**
  if *scheme* = StrangSplitting **and** β_{ok} **then**
    $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{Strang}(u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t, \beta_U, \beta_L)$ ▷ §2.7
  else
    $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{LieTrotter}(u_k, \ln U_k, \ln L_k, \Delta t)$ ▷ §2.6
  end if
  $u_{k+1} \leftarrow \text{ApplyBC}(u_{k+1})$ ▷ equation-specific, §2.5
  $\ln \tau_{k+1} \leftarrow \ln \tau(\ln U_{k+1}, \ln L_{k+1})$ ▷ (68), (89)
  $t_{k+1} \leftarrow t_k + \Delta t$
  if $k + 1 \geq 3$ **then**
    $(\beta_U, \beta_L, \beta_{\text{ok}}) \leftarrow \text{EstimateBeta}(\ln U_{k:k+1}, \ln L_{k:k+1}, \ln \tau_{k-2:k}, \Delta t, \beta_U, \beta_L)$ ▷ §2.8
  end if
  Store $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}, \ln \tau_{k+1}, t_{k+1}, \text{CFL}_k, \beta_U, \beta_L, \beta_{\text{ok}})$ ▷ (45)
  $k \leftarrow k + 1$
end while

The run parameter *scheme* selects the step function. Strang needs usable estimates of β_U, β_L, so the loop falls back to LieTrotter whenever β_{ok} is false. This is always the case for the first three steps $k = 0, 1, 2$: the first β update needs four stored states, so it first runs at the end of step $k = 2$. It is also the case for every step that follows a failed β update (§2.8), until a later update succeeds. The β update itself runs regardless of *scheme*; under *scheme* = LieTrotter the estimates are diagnostic only.

2.5 Substeps

Both splitting schemes are assembled from the substeps below. None of them applies the boundary treatment, which is handled only by the top-level loop.

Evolve$(u, \Delta t_k)$**.** Applies the physics operator $\Phi^{(\text{P})}_{\Delta t_k}$ of (9): it advances the non-scaling terms of (7) over Δt_k, leaving the scales $\ln U, \ln L$ untouched.

Scale$(u, \ln \lambda_U, \ln \lambda_L)$**.** Applies the scaling operator (19) with prescribed ratios λ_U, λ_L, returning $\lambda_U^{-1} u(\mathbf{x_n} \lambda_L)$ at each grid point, evaluated with the spectral interpolation of §1.3 and set to zero wherever $\mathbf{x_n} \lambda_L$ leaves the box (28). Here, and wherever the substeps are used, λ_U and λ_L denote the ratio of U, respectively L, after the substep to its value before it. Inside a Lie–Trotter step this is $\lambda_{U,k+1} = U_{k+1}/U_k$ of (18); inside a Strang step it is a ratio across a half step, $U_{k+1/2}/U_k$ or $U_{k+1}/U_{k+1/2}$ (§2.7).

Optional projection. With the run parameter *project* enabled, Scale ends by applying an equation-specific projection $\mathcal{P}$ onto the admissible fields, returning $\mathcal{P}\big(\lambda_U^{-1} u(\mathbf{x_n} \lambda_L)\big)$; with it disabled, or for an equation that defines no $\mathcal{P}$, the projection is the identity. The continuous operator (19) maps a divergence-free field to a divergence-free field, but its evaluation on the grid by interpolation and the out-of-box cutoff does not preserve this exactly. For Navier–Stokes, $\mathcal{P}$ is the Leray projection (162), which restores $\nabla \cdot \tilde{\mathbf{u}} = 0$ (§5.10); the 1D diffusion equation defines none. Because every scaling step runs through Scale, the projection follows the scaling step of LieTrotter and *both* scaling half steps of Strang. Inside ConstrainedScale it is applied after the ratios have been solved, so the returned field meets the targets F_1, F_2 only up to the

part the projection removes. The miss does not accumulate: the next ConstrainedScale measures the functionals of the projected field afresh and absorbs it into the next ratios.

ConstrainedScale$(u, \ln U, \ln L)$**.** Rescales u so that the result carries the fixed targets F_1, F_2 of (44), and advances the log-scale state accordingly:

1. **Functionals of the input,** $F_m^{\text{in}} = F_m[u]$, evaluated on the grid via (43).
2. **Ratios** $\ln \lambda_U, \ln \lambda_L$ from the linear system (30), in closed form. This works on the scalars F_m^{in} alone and does not touch the field.
3. **Rescale,** $u' = \textsc{Scale}(u, \ln \lambda_U, \ln \lambda_L)$.
4. **Return** u' together with the advanced state $\ln U + \ln \lambda_U$, $\ln L + \ln \lambda_L$.

No time increment enters ConstrainedScale: the ratios are fixed by the functionals of the field it is given, not by the elapsed time. The same substep therefore serves as the full scaling step of Lie–Trotter and as the closing scaling half step of Strang.

PredictScale$(i, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t, \beta_U, \beta_L)$**.** Predicts the ratios over the fraction i of step k from the power-law form (31) rather than from the constraints. It evaluates $\ln \Delta t_k^*$ from (41), with the constant-step weights (36), and returns $\big(G_U(i, \beta_U; k),\, G_L(i, \beta_L; k)\big)$ using (38). The same function G_X is inverted for β in §2.8. Both use $\ln A_X = 0$, which the step does not need (§1.5).

ApplyBC(u)**.** Imposes whatever boundary treatment the equation requires, after the full step has been assembled. Both equations treated here clamp the domain boundary to zero: the 1D diffusion equation its two edge grid points (§4.5), the Navier–Stokes equations the six faces of the box (§5.11). ApplyBC performs no projection; that is done only inside Scale. For Navier–Stokes the clamp is itself optional: with it disabled ApplyBC is the identity, and the far-field condition is left to the sponge forcing of §5.3, which acts inside Evolve.

2.6 LieTrotter: Lie–Trotter splitting

A Lie–Trotter step composes a full physics step with a full scaling step, $u_{k+1} = \Phi^{(\mathrm{S})}_{\Delta t_k}\big(\Phi^{(\mathrm{P})}_{\Delta t_k}(u_k)\big)$, with the scaling ratios fixed by the constraints. It uses no estimate of β_U, β_L, so it is valid from the first step. It is written for a general input state $(u, \ln U, \ln L)$ and step Δt_k rather than only a stored state $(u_k, \ln U_k, \ln L_k)$, so that Strang can call it on a half-stepped state.

Algorithm 2 Lie–Trotter step

```
function LieTrotter(u, ln U, ln L, Δt_k)
    u^(P) ← Evolve(u, Δt_k)                          ▷ §2.5
    return ConstrainedScale(u^(P), ln U, ln L)       ▷ §2.5
end function
```

2.7 Strang: Strang splitting

A Strang step is the symmetric composition

$$u_{k+1} = \Phi^{(\mathrm{S})}_{\Delta t_k/2}\Big(\Phi^{(\mathrm{P})}_{\Delta t_k}\big(\Phi^{(\mathrm{S})}_{\Delta t_k/2}(u_k)\big)\Big) \tag{46}$$

of a scaling half step, a full physics step, and a second scaling half step. The last two of these are exactly a Lie–Trotter step — a full physics step followed by a scaling step whose ratios come from the constraints — so Strang is a predicted scaling half step followed by a call to LieTrotter.

Algorithm 3 Strang step

function STRANG($u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t, \beta_U, \beta_L$)
  $\big(\ln(U_{k+1/2}/U_k),\ \ln(L_{k+1/2}/L_k)\big) \leftarrow$ PREDICTSCALE($\frac{1}{2}, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t, \beta_U, \beta_L$) ▷ (47)
  $u_{k+1/2} \leftarrow$ SCALE($u_k, \ln(U_{k+1/2}/U_k), \ln(L_{k+1/2}/L_k)$)
  $\ln U_{k+1/2} \leftarrow \ln U_k + \ln(U_{k+1/2}/U_k)$; $\ln L_{k+1/2} \leftarrow \ln L_k + \ln(L_{k+1/2}/L_k)$
  return LIETROTTER($u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2}, \Delta t_k$) ▷ §2.6
end function

No constraint fixes the state at $t_k + \Delta t_k/2$, so the first scaling half step takes its ratios from the current power-law estimates, via PREDICTSCALE with $i = 1/2$:

$$\ln \frac{U_{k+1/2}}{U_k} = G_U\big(\tfrac{1}{2}, \beta_U; k\big)\,, \qquad \ln \frac{L_{k+1/2}}{L_k} = G_L\big(\tfrac{1}{2}, \beta_L; k\big)\,. \tag{47}$$

The call to LIETROTTER then advances $u_{k+1/2}$ over the full Δt_k, and its CONSTRAINEDSCALE substep — the second scaling half step — sets $U_{k+1}/U_{k+1/2}$ and $L_{k+1}/L_{k+1/2}$ so that u_{k+1} satisfies the constraints. The half-step state $(u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2})$ is internal to the step and is not stored. The top-level loop calls STRANG only when β_{ok} is true (§2.4), which guarantees that β_U, β_L were fitted to the stored history at the end of the previous step and that $\ln \tau_{k-2}$ exists.

2.8 EstimateBeta: estimating β_U and β_L

After every step with $k+1 \geq 3$, regardless of *scheme*, β_U and β_L are re-estimated from the log-scale history. Four stored states are needed: $\ln X_k$ and $\ln X_{k+1}$ for the ratio, and $\ln \tau_{k-2:k}$ for Δt_k^*. The update inverts the function G_X of (38) with $i = 1$, the same function that PREDICTSCALE evaluates forward with $i = 1/2$:

$$G_U(1, \beta_U; k) = \ln U_{k+1} - \ln U_k, \tag{48}$$

and identically with $U \to L$ for β_L. The Δt_k^* inverted here is the one PREDICTSCALE used for that step's half step, so prediction and estimation are exact inverses on the same step. Since G_X carries $\ln A_X = 0$, the roots are the effective exponents $\tilde{\beta}_X$ of §1.5, which absorb the prefactor and tend to β_X as the scales grow; we write them β_X in the algorithm.

The sensitivity of β_X to an error in Δt_k^* is weak. Taking logarithms of (38) for a small step, $\ln(\ln X_{k+1} - \ln X_k) \approx \ln \beta_X + \ln \Delta t_k^* - \ln X_k/\beta_X$, so a relative error η in Δt_k^* moves β_X by $\delta\beta_X \approx \beta_X^2 \eta / \ln X_k$, damped as the scales grow.

Each of (48) and its L-counterpart is a single scalar equation in a single unknown (β_U, resp. β_L), solved by `brentq`; the implementer will need to choose a bracketing interval (e.g. centred on the previous β_U, β_L estimate — on $\beta_U^{(0)}, \beta_L^{(0)}$ for the first update — widened adaptively if it fails to bracket a root).

If both root-finds succeed, ESTIMATEBETA returns the new β_U, β_L with $\beta_{\text{ok}} = \text{true}$. If either fails to bracket a root even after widening, the failed estimate keeps its previous value (which seeds the next bracket), and ESTIMATEBETA returns $\beta_{\text{ok}} = \text{false}$, so the next step falls back to LIETROTTER. Before $k + 1 \geq 3$ no update is made: β_U, β_L stay at $\beta_U^{(0)}, \beta_L^{(0)}$ and β_{ok} stays false.

3 Post-processing

A completed run leaves behind the log-scale history $\ln U_k, \ln L_k$ and the clock history $\ln \tau_k$. From these we recover the parameters A_X, α_X, β_X of the ansatz (31). This is the offline, global counterpart of ESTIMATEBETA (§2.8), which fits an effective exponent one step at a time with $\ln A_X = 0$; here the whole history is used at once, which separates the prefactor from the exponent, and the fit also serves as a check on the run.

Write

$$\mathcal{L}_k = \ln L_k, \qquad \mathcal{U}_k = \ln U_k, \tag{49}$$

together with the per-step log-ratios $\ln \lambda_{X,k} = \ln X_k - \ln X_{k-1}$ of (18), which for a Strang step combine both half-step ratios. The stored scales are their cumulative sums,

$$\mathcal{L}_k = \mathcal{L}_0 + \sum_{i=1}^{k} \ln \lambda_{L,i}, \qquad \mathcal{U}_k = \mathcal{U}_0 + \sum_{i=1}^{k} \ln \lambda_{U,i}, \tag{50}$$

which grow only linearly in k; unlike L_k and U_k themselves, $\mathcal{L}_k$ and $\mathcal{U}_k$ never overflow.

The fitted range. The power law describes the run only once the initial transient has passed. We discard the record before a cutoff time t_{cut}, the same discard as in step 1 of §C.1, and fit over the steps $k_{\mathrm{c}} \le k \le K$, where $t_{k_{\mathrm{c}}} = t_{\mathrm{cut}}$ and K is the last step. The origin of physical time is placed at the cutoff,

$$t^*_{k_{\mathrm{c}}} = 0, \tag{51}$$

so that α_X is the offset of the power law at the start of the fitted record, $X_{k_{\mathrm{c}}} = A_X\, \alpha_X^{\beta_X}$.

3.1 The scaling rates under the ansatz

Differentiating the ansatz (31) with respect to t, and using the clock (3) to convert from t^*, the logarithmic rates (4) are

$$\sigma_X = \frac{1}{X}\frac{\mathrm{d}X}{\mathrm{d}t^*}\frac{\mathrm{d}t^*}{\mathrm{d}t} = \frac{\beta_X\, \tau}{t^* + \alpha_X}, \qquad X \in \{U, L\}, \tag{52}$$

independent of the prefactor. Inverting the ansatz to eliminate the unknown offset,

$$t^* + \alpha_X = \left(\frac{X}{A_X}\right)^{1/\beta_X}, \tag{53}$$

and substituting into (52) gives the rates in terms of the scales alone,

$$\sigma_X = \beta_X\, \tau \left(\frac{X}{A_X}\right)^{-1/\beta_X}. \tag{54}$$

The rate has the sign of β_X. Taking logarithms of its magnitude,

$$\ln|\sigma_{X,k}| = \ln|\beta_X| + \frac{\ln A_X}{\beta_X} + \ln \tau_k - \frac{\mathcal{X}_k}{\beta_X}, \qquad \mathcal{X}_k = \ln X_k. \tag{55}$$

Finally, the rates themselves are estimated directly from the log-ratios already in hand, without any finite difference of L_k or U_k themselves,

$$\sigma_{X,k} \approx \frac{\ln \lambda_{X,k+1}}{\Delta t_k}, \qquad X \in \{U, L\}, \tag{56}$$

The rates, not the log-ratios $\ln \lambda_{X,k+1}$ themselves, are also what the run's diagnostics plot, since a ratio scales with the step and the rates do not, so runs at different Δt compare directly.

3.2 Recovering A_X, α_X and β_X

Equation (55) is linear in $\mathcal{X}_k$ once the clock is divided out. Regressing

$$\ln|\sigma_{X,k}| - \ln \tau_k = b_X - \frac{1}{\beta_X}\,\mathcal{X}_k, \qquad b_X = \ln|\beta_X| + \frac{\ln A_X}{\beta_X}, \tag{57}$$

against $\mathcal{X}_k$ by least squares over the fitted range gives the slope $-1/\beta_X$ and the intercept b_X, and hence

$$\beta_X = -\frac{1}{\text{slope}}, \qquad \ln A_X = \beta_X\,(b_X - \ln|\beta_X|)\,. \tag{58}$$

The exponent is read off the slope alone and does not depend on the prefactor; the prefactor is whatever the intercept leaves over once β_X is known. The scatter of (57) about the fitted line is the first check on the ansatz: on a power law the points fall on a straight line, and a range that includes the initial transient, before the frame has locked on, bends it.

With A_X and β_X in hand, (53) gives the power-law time at each step,

$$\theta_{X,k} = \left(\frac{X_k}{A_X}\right)^{1/\beta_X} = \exp\!\left(\frac{\mathcal{X}_k - \ln A_X}{\beta_X}\right), \tag{59}$$

which should equal $t_k^* + \alpha_X$. The physical time is accumulated from the per-step increments of (34), with the constant-step weights (36) (the first two steps of the run, which lack a three-point history, take $\Delta t_i^* \approx \Delta t_i \tau_i$), from the origin (51),

$$t_k^* = \sum_{i=k_c}^{k-1} \Delta t_i^*, \qquad k \geq k_c. \tag{60}$$

In exact arithmetic $\theta_{X,k} - t_k^*$ is independent of k and equals α_X. In practice $\theta_{X,k}$ inherits a *relative* error from the fit — an error δ in $\ln A_X$ multiplies it by $e^{-\delta/\beta_X}$ — while t_k^* grows roughly exponentially in t on the logarithmic clock, so the absolute scatter of $\theta_{X,k} - t_k^*$ grows with $\theta_{X,k}$. We therefore take α_X as the least-squares fit of the relative misfit $(\theta_{X,k} - t_k^* - \alpha_X)/\theta_{X,k}$,

$$\alpha_X = \frac{\sum_k \theta_{X,k}^{-2}\,(\theta_{X,k} - t_k^*)}{\sum_k \theta_{X,k}^{-2}}, \tag{61}$$

summed over the fitted range. The weights concentrate it at the start of the range, where t_k^* is comparable to α_X and the offset is resolved at all. The root-mean-square of the relative misfit $(t_k^* + \alpha_X)/\theta_{X,k} - 1$ over the range is the second check on the ansatz: it measures how well the power law, with all three parameters fixed, reproduces the physical time the clock accumulates.

For the 1D diffusion equation the clock is $\tau = L^2/D$, and (55) specializes to

$$\ln \sigma_{L,k} = \ln \frac{\beta_L}{D} + \frac{\ln A_L}{\beta_L} + \left(2 - \frac{1}{\beta_L}\right) \mathcal{L}_k, \tag{62}$$

$$\ln |\sigma_{U,k}| = \ln \frac{|\beta_U|}{D} + \frac{\ln A_U}{\beta_U} + 2\,\mathcal{L}_k - \frac{\mathcal{U}_k}{\beta_U}, \tag{63}$$

so that for $X = L$ the regression (57) collapses onto a single variable, while for $X = U$ it is a two-variable fit in $\mathcal{L}_k$ and $\mathcal{U}_k$.

4 The 1D diffusion equation

The first instantiation of the framework is the one-dimensional diffusion equation, $d = 1$, with a single scalar component; we drop the component index throughout this section. It serves as a testbed: every operator the framework needs is available in closed form, so the only approximation in a run is the operator splitting itself.

4.1 Governing equation and clock

The dimensional equation is

$$\frac{\partial u^*}{\partial t^*} = D \frac{\partial^2 u^*}{\partial (x^*)^2}, \tag{64}$$

so that, in the notation of (1),

$$R^*(u^*) = D \frac{\partial^2 u^*}{\partial (x^*)^2}. \tag{65}$$

Under the rescaling (2) this term becomes

$$D\frac{\partial^2(uU)}{\partial(x^*)^2} = DU\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\frac{1}{L}\right)\frac{1}{L} = \frac{DU}{L^2}\frac{\partial^2 u}{\partial x^2}, \tag{66}$$

having used $\mathrm{d}x/\mathrm{d}x^* = 1/L$. The clock is then chosen to make the physical term of the lemma (7) carry unit coefficient,

$$\frac{\tau}{U}\cdot\frac{DU}{L^2} = 1 \quad\Longrightarrow\quad \tau = \frac{L^2}{D}, \tag{67}$$

which is the diffusive time scale of the current length scale, as expected. In log form, as required by (41),

$$\ln\tau_k = 2\ln L_k - \ln D. \tag{68}$$

Substituting (67) into the rescaling lemma (7) gives the equation the algorithm integrates,

$$\frac{\partial u}{\partial t} = \sigma_L\, x\frac{\partial u}{\partial x} - \sigma_U\, u + \frac{\partial^2 u}{\partial x^2}. \tag{69}$$

4.2 The physics step

Because the diffusion term in (69) has unit coefficient, the physics operator (9),

$$\Phi^{(\mathrm{P})}_{\Delta t}(u_k) = u_k + \int_{t_k}^{t_k+\Delta t}\frac{\partial^2 u}{\partial x^2}\,\mathrm{d}t, \tag{70}$$

has the closed-form spectral solution

$$\hat{u}^{(\mathrm{P})}_j = \hat{u}_j\, e^{-\kappa_j^2\Delta t}, \qquad \kappa_j = \frac{2\pi j}{H}, \quad j = 0,\ldots,\lfloor N/2\rfloor, \tag{71}$$

where $\hat{u}_j = \mathrm{RFFT}(u)_j$. This is an exact one-shot spectral propagator, not an approximate time-marching scheme, so no sub-stepping or implicit solve is needed, and the accuracy requirement on Evolve (§2.5) is met trivially at any order. Inverse-transforming gives the advanced field $u^{(\mathrm{P})}(x_n)$.

4.3 Moment constraints

The constraint functionals of §1.4 are taken to be the zeroth and the normalized second moment of the field,

$$F_1[u] = \int u\,\mathrm{d}x \equiv M_0, \qquad F_2[u] = \frac{1}{M_0}\int x^2 u\,\mathrm{d}x \equiv M_2, \tag{72}$$

where the M_0 in F_2 is the target value (44), a constant, not the zeroth moment of the field the functional is applied to. Thus M_0 measures the amount of the conserved quantity and M_2, a mean-square width, its spread. Writing $M_0^{\mathrm{in}}, M_2^{\mathrm{in}}$ for the moments of the field being scaled, and substituting $y = x\,\lambda_L$ in each integral, the scaled field carries

$$\int \frac{1}{\lambda_U}u(x\,\lambda_L)\,\mathrm{d}x = \frac{1}{\lambda_U\lambda_L}\,M_0^{\mathrm{in}}, \tag{73}$$

$$\int x^2\,\frac{1}{\lambda_U}u(x\,\lambda_L)\,\mathrm{d}x = \frac{1}{\lambda_U\lambda_L^3}\,M_2^{\mathrm{in}}, \tag{74}$$

so the homogeneity exponents of (29) are

$$(p_1,q_1) = (1,1), \qquad (p_2,q_2) = (1,3), \tag{75}$$

with $p_1q_2 - p_2q_1 = 2 \neq 0$, so the constraint system (30) is uniquely solvable: the two moments do respond independently to amplitude and to length rescaling. Solving it,

$$\lambda_L = \sqrt{\frac{M_0\,M_2^{\mathrm{in}}}{M_0^{\mathrm{in}}\,M_2}}, \tag{76}$$

$$\lambda_U = \frac{M_0^{\mathrm{in}}}{M_0\,\lambda_L} = \left(\frac{M_0^{\mathrm{in}}}{M_0}\right)^{3/2} \sqrt{\frac{M_2}{M_2^{\mathrm{in}}}}. \tag{77}$$

Because the normalization of F_2 is a constant, these ratios are the same as for the raw second moment $\int x^2 u\,\mathrm{d}x$; written in terms of the raw moments, M_0 cancels from M_2^{in}/M_2.

4.4 Spectral interpolation on the real transform

For $d = 1$ and a real-valued field, the generic interpolation formula (27) is realized on the non-negative frequency components of a real Fast Fourier Transform (RFFT):

$$\begin{aligned}\Phi^{(\mathrm{S})}_{\Delta t}(u_k)(x) = \frac{1}{\lambda_{U,k+1}}\Bigg\{ \hat{u}_0^{(k)} + 2\sum_{j=1}^{N/2-1} (-1)^j \Bigg[\mathrm{Re}\big(\hat{u}_j^{(k)}\big)\cos\left(\frac{2\pi j \lambda_{L,k+1} x}{H}\right) \\ - \mathrm{Im}\big(\hat{u}_j^{(k)}\big)\sin\left(\frac{2\pi j \lambda_{L,k+1} x}{H}\right)\Bigg] \\ + (-1)^{N/2}\,\hat{u}_{N/2}^{(k)}\cos\left(\frac{\pi N \lambda_{L,k+1} x}{H}\right)\Bigg\},\end{aligned} \tag{78}$$

where $\hat{u}_j^{(k)} = \frac{1}{N}\mathrm{RFFT}(u_k)_j$ for $j = 0, 1, \ldots, N/2$, and the Nyquist term at $j = N/2$ is included only if N is even. The factor of 2 accounts for the Hermitian conjugate modes omitted by the real transform. (78) holds for $|\lambda_{L,k+1} x| \le H/2$; outside that the cutoff (28) sets $\Phi^{(\mathrm{S})}_{\Delta t}(u_k)(x) = 0$.

The alternating sign $(-1)^j$ is the phase-origin correction, and it is not optional. The transform phases its coefficients to the sample index n, whose origin $n = 0$ sits at $x = -H/2$ on the centred grid (42) rather than at $x = 0$. Since the scaling operator (19) dilates about the centre of the domain, each mode must be evaluated at $\lambda_{L,k+1} x + H/2$ — dilate about $x = 0$, then shift to the transform's phase origin — and not at $\lambda_{L,k+1}(x + H/2)$, which dilates the shift along with the coordinate and agrees with it only at $\lambda_{L,k+1} = 1$. As $H/2$ is half a period, that shift contributes the constant phase $\exp(\mathrm{i}\kappa_j H/2) = (-1)^j$. Dropping it altogether returns $u_k(\lambda_{L,k+1} x - H/2)$, the wanted profile displaced by half the domain, rather than the $u_k(\lambda_{L,k+1} x)$ that (19) calls for. The same factor carries to each axis in three dimensions (§5.10).

4.5 Boundary treatment

The substep ApplyBC of §2.5 imposes a homogeneous Dirichlet condition by clamping the two edge grid points of the centred domain (42) after each completed step,

$$u_{k+1}(x_0) \leftarrow 0, \qquad u_{k+1}(x_{N-1}) \leftarrow 0. \tag{79}$$

On the centred grid $x \in [-H/2, H/2)$ these are two distinct grid points, not periodic images of one another.

Open. This sits in tension with the rest of the discretization: the spectral propagator (71) presumes a periodic field, whereas (79) imposes a Dirichlet condition. The spectral interpolation (78) no longer does at the edges: with the cutoff (28) it treats everything beyond the box as zero, which is consistent with the clamp, but it is still a periodic interpolant inside the box. The two are consistent only so long as u is negligible near the domain edges, which requires the initial condition to be localized well inside the domain and the run to be stopped before the profile spreads to the boundary. The scaling step works against this — it is what keeps the profile from spreading in the self-similar frame — but the condition is an assumption of the scheme rather than a consequence of it, and is not currently monitored.

5 The Navier–Stokes equations

The second instantiation is the incompressible Navier–Stokes equations, with $d = 3$ and u_i the velocity field.

5.1 Governing equations and clock

The equations are solved as a large-eddy simulation (LES). Let $\mathcal{G}_{\Delta^*}$ denote a spatial filter of width Δ^*. A tilde marks a quantity of the filtered (resolved) field, and the same symbol without a tilde the corresponding quantity of the full field: $\tilde{u}_i^* = \mathcal{G}_{\Delta^*}(u_i^*)$ and $\tilde{p}^*$ are the filtered velocity and pressure, while u_i^* and p^* are the full ones. For a quantity derived from the velocity, the tilde means it is evaluated from $\tilde{u}_i^*$, not that it is filtered; thus $\frac{1}{2}\tilde{u}_i^*\tilde{u}_i^*$ is not $\mathcal{G}_{\Delta^*}(\frac{1}{2}u_i^*u_i^*)$. The effect of the unresolved scales enters through the kinematic subgrid-scale stress

$$\tau_{ij}^* = \mathcal{G}_{\Delta^*}(u_i^*u_j^*) - \tilde{u}_i^*\tilde{u}_j^*, \tag{80}$$

which, like the eddy viscosity below, exists only in the LES and carries no tilde. The dimensional filtered equations are

$$\frac{\partial \tilde{u}_i^*}{\partial t^*} + \tilde{u}_j^*\frac{\partial \tilde{u}_i^*}{\partial x_j^*} = -\frac{1}{\rho}\frac{\partial \tilde{p}^*}{\partial x_i^*} + \nu\frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial \tau_{ij}^*}{\partial x_j^*}, \tag{81}$$

together with the continuity equation,

$$\frac{\partial \tilde{u}_i^*}{\partial x_i^*} = 0, \tag{82}$$

so that, in the notation of (1),

$$R_i^*(\tilde{u}^*) = -\tilde{u}_j^*\frac{\partial \tilde{u}_i^*}{\partial x_j^*} - \frac{1}{\rho}\frac{\partial \tilde{p}^*}{\partial x_i^*} + \nu\frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial \tau_{ij}^*}{\partial x_j^*}. \tag{83}$$

In addition to the rescaling (2) of $\tilde{u}_i$ and x_j, the pressure is nondimensionalized by the dynamic pressure,

$$\tilde{p} = \frac{\tilde{p}^*}{\rho U^2}, \tag{84}$$

and the subgrid-scale stress, which carries units of velocity squared, by U^2,

$$\tau_{ij} = \frac{\tau_{ij}^*}{U^2}. \tag{85}$$

With the filter width nondimensionalized as $\Delta = \Delta^*/L$, the definitions carry over unchanged, $\tilde{u}_i = \mathcal{G}_\Delta(u_i)$ and $\tau_{ij} = \mathcal{G}_\Delta(u_iu_j) - \tilde{u}_i\tilde{u}_j$. Term by term, the convective, pressure, and subgrid-scale terms scale as U^2/L,

$$\tilde{u}_j^*\frac{\partial \tilde{u}_i^*}{\partial x_j^*} = \frac{U^2}{L}\tilde{u}_j\frac{\partial \tilde{u}_i}{\partial x_j}, \qquad \frac{1}{\rho}\frac{\partial \tilde{p}^*}{\partial x_i^*} = \frac{U^2}{L}\frac{\partial \tilde{p}}{\partial x_i}, \qquad \frac{\partial \tau_{ij}^*}{\partial x_j^*} = \frac{U^2}{L}\frac{\partial \tau_{ij}}{\partial x_j}, \tag{86}$$

while the viscous term scales as $\nu U/L^2$,

$$\nu\frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*} = \frac{\nu U}{L^2}\frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j}. \tag{87}$$

The clock is chosen as the inverse of the advection timescale

$$\tau = \frac{L}{U}. \tag{88}$$

In log form, as required by (41),

$$\ln \tau_k = \ln L_k - \ln U_k. \tag{89}$$

Applying $\tau/U = L/U^2$ to (83) and introducing the instantaneous Reynolds number

$$Re = \frac{UL}{\nu}, \tag{90}$$

the rescaling lemma (7) gives the nondimensional momentum equation

$$\frac{\partial \tilde{u}_i}{\partial t} = \sigma_L\, x_j\frac{\partial \tilde{u}_i}{\partial x_j} - \sigma_U\, \tilde{u}_i - \tilde{u}_j\frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} + \frac{1}{Re}\frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j}, \tag{91}$$

and the continuity equation is unchanged by the rescaling,

$$\frac{\partial \tilde{u}_i}{\partial x_i} = 0. \tag{92}$$

5.2 Subgrid stress model

The subgrid-scale stress τ^*_{ij} of (81) is closed with an eddy-viscosity model. Its deviatoric part is taken proportional to the resolved strain rate,

$$\tau^*_{ij} - \frac{1}{3}\tau^*_{kk}\delta_{ij} = -2\nu^*_{\mathrm{t}}\tilde{S}^*_{ij}, \qquad \tilde{S}^*_{ij} = \frac{1}{2}\left(\frac{\partial \tilde{u}^*_i}{\partial x^*_j} + \frac{\partial \tilde{u}^*_j}{\partial x^*_i}\right), \tag{93}$$

with the isotropic part $\frac{1}{3}\tau^*_{kk}\delta_{ij}$ absorbed into the pressure. The eddy viscosity ν^*_{t} carries the units of UL, so it is nondimensionalized as $\nu_{\mathrm{t}} = \nu^*_{\mathrm{t}}/(UL)$. Noting that $\tilde{S}^*_{ij} = (U/L)\,\tilde{S}_{ij}$, the nondimensional subgrid stress (85) is

$$\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij} = -2\nu_{\mathrm{t}}\tilde{S}_{ij}. \tag{94}$$

The models below differ only in the choice of ν^*_{t}.

5.2.1 Smagorinsky model

The standard Smagorinsky model takes the eddy viscosity

$$\nu^*_{\mathrm{t}} = (C_s\Delta^*)^2|\tilde{S}^*|, \qquad |\tilde{S}^*| = \sqrt{2\tilde{S}^*_{ij}\tilde{S}^*_{ij}}, \tag{95}$$

where C_s is the Smagorinsky constant and Δ^* is the filter width. With the nondimensional filter width $\Delta = \Delta^*/L$ taken equal to the grid spacing, $\Delta = \Delta x$, the eddy viscosity becomes

$$\nu^*_{\mathrm{t}} = UL\,(C_s\Delta)^2|\tilde{S}|, \qquad \nu_{\mathrm{t}} = (C_s\Delta)^2|\tilde{S}|. \tag{96}$$

5.2.2 Self-similar constant eddy viscosity

The second model keeps the dependence on the filter width of (95) but replaces the local strain rate $|\tilde{S}^*|$ by the strain-rate scale U/L of the self-similar frame,

$$\nu^*_{\mathrm{t}} = C_\nu(\Delta^*)^2\frac{U}{L}, \qquad \nu_{\mathrm{t}} = C_\nu\Delta^2, \tag{97}$$

with C_ν a constant and $\Delta = \Delta^*/L = \Delta x$ as in the Smagorinsky model. The eddy viscosity is uniform in space, so with (82) the subgrid stress term reduces to a Laplacian,

$$-\frac{\partial \tau^*_{ij}}{\partial x^*_j} = \nu^*_{\mathrm{t}}\frac{\partial^2 \tilde{u}^*_i}{\partial x^*_j \partial x^*_j}, \qquad -\frac{\partial \tau_{ij}}{\partial x_j} = C_\nu\Delta^2\frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j}. \tag{98}$$

In the self-similar frame the model is therefore a constant viscosity, of the same form as the molecular term of (91) with an effective Reynolds number $Re_{\mathrm{t}} = 1/(C_\nu\Delta^2)$. Unlike Re, which changes with UL as the flow evolves, Re_{t} is fixed for the run, while in physical variables, where $\Delta^* = L\Delta$ grows with the flow, ν^*_{t} follows the product UL.

The momentum equation

In the limit of infinite Reynolds number, $1/Re = 0$, and with either model the momentum equation (91) becomes

$$\frac{\partial \tilde{u}_i}{\partial t} = \sigma_L\, x_j\frac{\partial \tilde{u}_i}{\partial x_j} - \sigma_U\,\tilde{u}_i - \tilde{u}_j\frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} + \frac{\partial}{\partial x_j}\left[2\nu_{\mathrm{t}}\,\tilde{S}_{ij}\right], \tag{99}$$

where $\tilde{p}$ now denotes the modified pressure that includes $\frac{1}{3}\tau_{kk}$, and ν_{t} is given by (96) or (97). For the self-similar constant eddy viscosity the last term is $C_\nu\Delta^2\,\partial^2\tilde{u}_i/\partial x_j\partial x_j$, by (98).

5.3 Sponge forcing

The true boundary conditions we would like to enforce is $\tilde{u}_i = 0$ at the edge of the domain. However, we are using a pseudo-spectral solver so this boundary condition cannot be directly enforced. To solve this we introduce a fringe region at the edge of the domain [1], which introduces a forcing term to the right hand side of the Navier–Stokes equation

$$f_i = -W(r)\,\tilde{u}_i, \tag{100}$$

where $r = \sqrt{x_i x_i}$ is the distance from the origin and $W(r)$ is a weighting function that increases with increasing r.

The weighting function is

$$W(r) = W_{\max}\, S\!\left(\frac{r - r_0}{r_1 - r_0}\right), \tag{101}$$

where the smooth step S is

$$S(\xi) = \begin{cases} 0, & \xi \le 0, \\ \dfrac{1}{1 + \exp\!\left(\dfrac{1}{\xi - 1} + \dfrac{1}{\xi}\right)}, & 0 < \xi < 1, \\ 1, & \xi \ge 1. \end{cases} \tag{102}$$

The function S is C^∞ on the whole real line. Every derivative of it vanishes at $\xi = 0$ and at $\xi = 1$, it rises monotonically between them, and it is antisymmetric about its midpoint, $S(1-\xi) = 1 - S(\xi)$, so that $S(1/2) = 1/2$. The weight (101) is therefore exactly zero in the sphere $r \le r_0$, rises smoothly across the shell $r_0 < r < r_1$, and is exactly $W_{\max}$ for $r \ge r_1$. Adding (100) to (99), the momentum equation actually solved is

$$\frac{\partial \tilde{u}_i}{\partial t} = \sigma_L\, x_j \frac{\partial \tilde{u}_i}{\partial x_j} - \sigma_U\, \tilde{u}_i - \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} + \frac{\partial}{\partial x_j}\left[2\nu_{\mathrm{t}}\, \tilde{S}_{ij}\right] - W(r)\,\tilde{u}_i. \tag{103}$$

5.4 Fourier representation

Using the Fourier convention (20), the pressure gradient becomes

$$\widehat{\frac{\partial \tilde{p}}{\partial x_i}} = \mathrm{i}\kappa_i \hat{\tilde{p}}. \tag{104}$$

The convective term

$$\hat{N}_i = \widehat{\tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j}}, \tag{105}$$

will be evaluated pseudo-spectrally. How the subgrid stress term

$$\hat{G}_i = \frac{\partial}{\partial x_j} \widehat{\left[2\nu_{\mathrm{t}} \tilde{S}_{ij}\right]} \tag{106}$$

is evaluated depends on the model of §5.2. In both cases the strain rate is available spectrally,

$$\hat{\tilde{S}}_{ij} = \frac{\mathrm{i}}{2}\left(\kappa_j \hat{\tilde{u}}_i + \kappa_i \hat{\tilde{u}}_j\right). \tag{107}$$

Smagorinsky model. The eddy viscosity (96) depends on the field, so $\hat{G}_i$ is evaluated pseudo-spectrally: $\tilde{S}_{ij}$ is transformed to physical space, the products $|\tilde{S}|$, $\nu_{\mathrm{t}} = (C_s\Delta)^2|\tilde{S}|$ and $T_{ij} = 2\nu_{\mathrm{t}}\tilde{S}_{ij}$ are formed there, and the divergence is taken spectrally,

$$\hat{G}_i = \mathrm{i}\kappa_j \hat{T}_{ij}. \tag{108}$$

Since $|\tilde{S}|$ is not a polynomial in $\tilde{u}$, $\hat{G}_i$ carries aliasing that the 2/3 rule does not remove entirely.

Self-similar constant eddy viscosity. The eddy viscosity (97) is uniform, and by (98)

$$\hat{G}_i = -C_\nu \Delta^2 \kappa^2 \hat{\tilde{u}}_i. \tag{109}$$

This is exact and diagonal in $\boldsymbol{\kappa}$: no product is formed in physical space, so $\hat{G}_i$ needs no transforms, carries no aliasing, and stays in the retained band. By (112) it is also solenoidal, $\kappa_i \hat{G}_i = 0$, so it contributes nothing to the pressure below and passes through the projection (118) unchanged.

Sponge forcing. The sponge forcing (100) is formed as a product in physical space and transformed,

$$\hat{F}_i = \widehat{W \tilde{u}_i}, \tag{110}$$

with W tabulated once on the grid (42). Because W is smooth but not band-limited, $\hat{F}_i$ also carries content beyond the retained band, which the 2/3 mask removes along with that of $\hat{N}_i$ and, under the Smagorinsky model, $\hat{G}_i$. The scaling terms transform as in (23) with $d = 3$, so the transform of (103) is

$$\frac{\partial \hat{\tilde{u}}_i}{\partial t} = -\sigma_L \left(\kappa_j \frac{\partial \hat{\tilde{u}}_i}{\partial \kappa_j} + 3\, \hat{\tilde{u}}_i \right) - \sigma_U\, \hat{\tilde{u}}_i - \hat{N}_i - \mathrm{i}\kappa_i \hat{\tilde{p}} + \hat{G}_i - \hat{F}_i, \tag{111}$$

and the transform of the continuity equation (92) is

$$\kappa_i \hat{\tilde{u}}_i = 0. \tag{112}$$

5.5 Pressure and the solenoidal projection

Contracting (111) with κ_i and using the continuity constraint gives

$$0 = -\sigma_L \left(\kappa_i \kappa_j \frac{\partial \hat{\tilde{u}}_i}{\partial \kappa_j} + d\, \kappa_i \hat{\tilde{u}}_i \right) - \sigma_U\, \kappa_i \hat{\tilde{u}}_i - \kappa_i \hat{N}_i - \mathrm{i}\kappa^2 \hat{\tilde{p}} + \kappa_i \hat{G}_i - \kappa_i \hat{F}_i. \tag{113}$$

The remaining scaling contribution also vanishes: differentiating (112) with respect to κ_j gives

$$\frac{\partial}{\partial \kappa_j} \left(\kappa_i \hat{\tilde{u}}_i \right) = \hat{\tilde{u}}_j + \kappa_i \frac{\partial \hat{\tilde{u}}_i}{\partial \kappa_j} = 0 \quad \Longrightarrow \quad \kappa_i \frac{\partial \hat{\tilde{u}}_i}{\partial \kappa_j} = -\hat{\tilde{u}}_j, \tag{114}$$

so that

$$\kappa_i \kappa_j \partial \hat{\tilde{u}}_i / \partial \kappa_j = -\kappa_j \hat{\tilde{u}}_j = 0. \tag{115}$$

Therefore, the equation for pressure is

$$\mathrm{i}\kappa^2 \hat{\tilde{p}} = -\kappa_i \left(\hat{N}_i - \hat{G}_i + \hat{F}_i \right) \quad \Longrightarrow \quad \hat{\tilde{p}} = \mathrm{i}\, \frac{\kappa_l \left(\hat{N}_l - \hat{G}_l + \hat{F}_l \right)}{\kappa^2}, \qquad \kappa \neq 0, \tag{116}$$

with $\hat{\tilde{p}}$ at $\kappa = 0$ fixed arbitrarily, since only its gradient enters the momentum equation. Substituting (116) into (111), the pressure term becomes

$$-\mathrm{i}\kappa_i \hat{\tilde{p}} = \frac{\kappa_i \kappa_l}{\kappa^2} \left(\hat{N}_l - \hat{G}_l + \hat{F}_l \right), \tag{117}$$

which combines with $-\hat{N}_i + \hat{G}_i - \hat{F}_i$ to give the solenoidal projection of the convective, subgrid stress, and sponge terms together. Defining the projection tensor

$$P_{il}(\boldsymbol{\kappa}) = \delta_{il} - \frac{\kappa_i \kappa_l}{\kappa^2}, \tag{118}$$

the evolution equation in fourier space is (7),

$$\frac{\partial \hat{\tilde{u}}_i}{\partial t} = \underbrace{-\sigma_L \left(\kappa_j \frac{\partial \hat{\tilde{u}}_i}{\partial \kappa_j} + 3\, \hat{\tilde{u}}_i \right) - \sigma_U\, \hat{\tilde{u}}_i}_{\text{scaling terms}} \underbrace{- P_{il}(\boldsymbol{\kappa}) \left(\hat{N}_l - \hat{G}_l + \hat{F}_l \right)}_{\text{physical terms}}. \tag{119}$$

5.6 The physics step

The physics operator (9) is the flow of the bracketed physical terms of (119),

$$\Phi^{(\mathrm{P})}_{\Delta t}\big(\hat{\tilde{u}}_i(t)\big) = \hat{\tilde{u}}_i(t) - \int_t^{t+\Delta t} P_{il}(\boldsymbol{\kappa}) \left(\hat{N}_l - \hat{G}_l + \hat{F}_l\right) \mathrm{d}t. \tag{120}$$

We take $\Phi^{(\mathrm{P})}$ to be a single step of the classical fourth-order Runge–Kutta method (RK4) applied to the full physical right-hand side, the subgrid stress term $\hat{G}_i$ and the sponge forcing $\hat{F}_i$ marched together with the convection. It is set out in Appendix A, together with the stability limits it places on Δt. The advective limit depends on the state and is monitored through the realized CFL number (45) of §2.3. The sponge limit and, under the self-similar constant eddy viscosity, the diffusive limit are fixed by the run parameters. None of the three is enforced: the constant step Δt must be chosen to satisfy them.

5.7 Post-processing

We report two pointwise statistics as functions of the distance from the origin: the trace of the dissipation tensor and the kinetic energy (§5.7.2). Each is a dimensional prefactor built from U and L times a nondimensional quantity computed from the self-similar field, so a single scaling argument, given in §5.7.1, covers both. §5.7.3 then derives the budget of the dissipation. This subsection comes before the constraint functionals of §5.8 because their integrand is the resolved kinetic energy defined here, and the subgrid estimate defined here measures the energy they neglect.

5.7.1 Scaling of point statistics

Let Q^* be a pointwise quantity that nondimensionalizes as

$$Q^* = U^{m_U} L^{m_L} Q, \tag{121}$$

where the nondimensional Q is evaluated from the self-similar fields. By the rescaling (2), a fixed point $\mathbf{x}$ of the self-similar frame corresponds to the moving physical point $\mathbf{x}^* = L\,\mathbf{x}$. Evaluating at $\mathbf{x}$ on the grid (42) therefore requires no interpolation. Substituting the power-law ansatz (31) for U and L,

$$Q^*(L\mathbf{x}, t^*) = A_U^{m_U} A_L^{m_L}\, (t^* + \alpha_U)^{m_U \beta_U} (t^* + \alpha_L)^{m_L \beta_L}\, Q(\mathbf{x}, t). \tag{122}$$

The origin, $\mathbf{x}^* = \mathbf{x} = 0$, is the only point fixed in both frames ($n = N/2$ for even N). Elsewhere (122) relates the physical field to the self-similar one along the ray $\mathbf{x}^* \propto L$, not at a fixed physical position.

Scaling of the ensemble average. Let $\overline{\cdot}$ denote the average over an ensemble of initial conditions. We make two assumptions:

1. The parameters $A_U, \alpha_U, \beta_U, A_L, \alpha_L, \beta_L$ are the same for every member of the ensemble, i.e. $U(t^*)$ and $L(t^*)$ do not depend on the initial condition.
2. In the self-similar frame the flow is statistically stationary, so that $\overline{Q(\mathbf{x}, t)}$ is independent of t.

Then

$$\overline{Q^*}(L\mathbf{x}, t^*) = A_U^{m_U} A_L^{m_L}\, (t^* + \alpha_U)^{m_U \beta_U} (t^* + \alpha_L)^{m_L \beta_L}\, \overline{Q}(\mathbf{x}). \tag{123}$$

Isotropy. The governing equations, the sponge forcing (100) (whose weight $W(r)$ depends only on r), and the ensemble of initial conditions are all invariant under rotations about the origin. We therefore assume that the ensemble statistics are too. A scalar statistic, such as the trace of the dissipation tensor or the kinetic energy, then depends only on $r = \sqrt{x_i x_i}$, $\overline{Q}(\mathbf{x}) = \overline{Q}(r)$, and (123) becomes

$$\overline{Q^*}(r^*, t^*) = A_U^{m_U} A_L^{m_L}\, (t^* + \alpha_U)^{m_U \beta_U} (t^* + \alpha_L)^{m_L \beta_L}\, \overline{Q}(r^*/L). \tag{124}$$

The flow is not homogeneous, so the statistics do vary with r; we assume only that they do not vary with direction. The periodic cubic domain breaks this symmetry, but only near its boundary, where the sponge damps the field (§5.3).

Every point on the sphere $|\mathbf{x}| = r$ is then a sample of the same statistic, so we can average over that sphere within a single snapshot. This is the spherical mean,

$$\langle Q \rangle_r(t) = \frac{1}{4\pi} \oint_{|\hat{\mathbf{n}}|=1} Q(r\hat{\mathbf{n}}, t)\, \mathrm{d}\Omega(\hat{\mathbf{n}}). \tag{125}$$

The sphere does not pass through grid points, so (125) is evaluated on the spectral interpolant of Q, where it reduces to a weighted sum over the Fourier modes (Appendix B). At $r = 0$ it is the point value, $\langle Q \rangle_0 = Q(0, t)$.

If the system is also ergodic, the ensemble average equals a time average,

$$\overline{Q}(r) = \frac{1}{T} \int_0^T \langle Q \rangle_r(t)\, \mathrm{d}t, \tag{126}$$

and a single long simulation suffices. In practice T is finite and the initial transient is discarded, so (126) only estimates $\overline{Q}(r)$; Appendix C describes how its statistical error is computed. The spherical mean reduces this error compared with a single point. The reduction is larger at larger r, where the sphere spans more correlation lengths, and absent at $r = 0$.

5.7.2 Dissipation and kinetic energy

The dissipation tensor and its trace. As in §6.1, the ensemble-mean velocity vanishes, so u_i^* is the fluctuating velocity. The evolution equation for the Reynolds stress tensor $\overline{u_i^* u_j^*}$, with $\overline{\cdot}$ the ensemble average of §5.7.1, contains the sink $-\varepsilon_{ij}^*$, where

$$\varepsilon_{ij}^* = 2\nu\, \overline{\frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_j^*}{\partial x_k^*}} \tag{127}$$

is the dissipation tensor [2]. We report its trace,

$$\epsilon^* = \varepsilon_{ii}^* = 2\nu\, \overline{\frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_i^*}{\partial x_k^*}}, \tag{128}$$

which is the sink in the equation for $\overline{u_i^* u_i^*} = 2\overline{k^*}$. It is twice the pseudo-dissipation. It differs from twice the dissipation rate $2\nu\, \overline{S_{ij}^* S_{ij}^*}$ by a divergence, which vanishes in homogeneous turbulence but not here.

Estimate from the filtered field. The LES does not resolve the gradients in (128). As for the kinetic energy below, we therefore estimate the filtered integrand, $\mathcal{G}_{\Delta^*}$ of it. We split it into a resolved and a subgrid part,

$$\mathcal{G}_{\Delta^*}\left(2\nu\, \frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_j^*}{\partial x_k^*}\right) = 2\nu\, \frac{\partial \tilde{u}_i^*}{\partial x_k^*} \frac{\partial \tilde{u}_j^*}{\partial x_k^*} + 2\nu \left[\mathcal{G}_{\Delta^*}\left(\frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_j^*}{\partial x_k^*}\right) - \frac{\partial \tilde{u}_i^*}{\partial x_k^*} \frac{\partial \tilde{u}_j^*}{\partial x_k^*}\right], \tag{129}$$

where the bracket is the unclosed subgrid gradient correlation. The eddy-viscosity closure (93) treats the subgrid scales as an additional viscosity ν_t^* acting on the resolved field. For consistency we close the bracket the same way, by gradient transport,

$$\mathcal{G}_{\Delta^*}\left(\frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_j^*}{\partial x_k^*}\right) \approx \left(1 + \frac{\nu_\mathrm{t}^*}{\nu}\right) \frac{\partial \tilde{u}_i^*}{\partial x_k^*} \frac{\partial \tilde{u}_j^*}{\partial x_k^*}. \tag{130}$$

The bracket in (129) then becomes $(\nu_\mathrm{t}^*/\nu)\,(\partial \tilde{u}_i^*/\partial x_k^*)(\partial \tilde{u}_j^*/\partial x_k^*)$, and

$$\mathcal{G}_{\Delta^*}\left(2\nu\, \frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_j^*}{\partial x_k^*}\right) \approx 2\nu\, \frac{\partial \tilde{u}_i^*}{\partial x_k^*} \frac{\partial \tilde{u}_j^*}{\partial x_k^*} + 2\nu_\mathrm{t}^*\, \frac{\partial \tilde{u}_i^*}{\partial x_k^*} \frac{\partial \tilde{u}_j^*}{\partial x_k^*} = 2\,(\nu + \nu_\mathrm{t}^*)\, \frac{\partial \tilde{u}_i^*}{\partial x_k^*} \frac{\partial \tilde{u}_j^*}{\partial x_k^*}. \tag{131}$$

Taking the trace gives the pointwise estimate of the integrand of (128),

$$\epsilon^* = 2\,(\nu + \nu_\mathrm{t}^*)\, \frac{\partial \tilde{u}_i^*}{\partial x_k^*} \frac{\partial \tilde{u}_i^*}{\partial x_k^*}. \tag{132}$$

Since the filter commutes with the ensemble average, the average of (132) estimates (128). With $\nu_{\mathrm{t}}^* = UL\,\nu_{\mathrm{t}}$, $\nu = UL/Re$, and $\partial\tilde{u}_i^*/\partial x_k^* = (U/L)\,\partial\tilde{u}_i/\partial x_k$,

$$\epsilon^* = \frac{U^3}{L}\,\epsilon, \qquad \epsilon = 2\left(\frac{1}{Re} + \nu_{\mathrm{t}}\right)\frac{\partial\tilde{u}_i}{\partial x_k}\frac{\partial\tilde{u}_i}{\partial x_k}, \tag{133}$$

which is of the form (121) with $m_U = 3$, $m_L = -1$. In the limit $1/Re = 0$ the resolved part vanishes and only the subgrid part remains. By (96) and (97),

$$\epsilon = 2\nu_{\mathrm{t}}\frac{\partial\tilde{u}_i}{\partial x_k}\frac{\partial\tilde{u}_i}{\partial x_k} = \begin{cases} 2(C_s\Delta)^2|\tilde{S}|\,\dfrac{\partial\tilde{u}_i}{\partial x_k}\dfrac{\partial\tilde{u}_i}{\partial x_k}, & \text{Smagorinsky model,} \\ 2C_\nu\Delta^2\,\dfrac{\partial\tilde{u}_i}{\partial x_k}\dfrac{\partial\tilde{u}_i}{\partial x_k}, & \text{self-similar constant eddy viscosity.} \end{cases} \tag{134}$$

Kinetic energy. The resolved kinetic energy is

$$\tilde{k}^* = U^2\,\tilde{k}, \qquad \tilde{k} = \frac{1}{2}\tilde{u}_i\tilde{u}_i. \tag{135}$$

Contracting (80) gives the filtered kinetic energy of the full field [2],

$$\mathcal{G}_\Delta(k) = \tilde{k} + k_{\mathrm{sgs}}, \qquad k_{\mathrm{sgs}} = \frac{1}{2}\tau_{kk}. \tag{136}$$

The isotropic part of τ_{ij} is absorbed into the pressure, so the solver never computes k_{sgs} and we must estimate it. It enters neither the integration nor the constraints (§5.8); we use it only as a diagnostic of how much energy the resolved field misses. For both subgrid stress models we estimate it with the Smagorinsky eddy viscosity (96), so that the estimate is the same function of the resolved field in either case. Equating the production $\nu_{\mathrm{t}}|\tilde{S}|^2 = C_s^2\Delta^2|\tilde{S}|^3$ to the dissipation $C_\epsilon k_{\mathrm{sgs}}^{3/2}/\Delta$, with C_ϵ a constant of order one, gives

$$k_{\mathrm{sgs}} = \left(\frac{C_s^2}{C_\epsilon}\right)^{2/3}\Delta^2|\tilde{S}|^2, \tag{137}$$

with $k_{\mathrm{sgs}}^* = U^2 k_{\mathrm{sgs}}$.

Radial profiles. The dissipation (134), the resolved kinetic energy (135), and the subgrid kinetic energy (137) are scalars. By the isotropy assumption of §5.7.1 we report each as a radial profile, $\overline{\epsilon}(r)$, $\overline{\tilde{k}}(r)$, and $\overline{k_{\mathrm{sgs}}}(r)$, estimated by the spherical mean (125) followed by the time average (126). The physical profiles follow from (124).

5.7.3 ϵ budget

This subsection derives the evolution equation of the mean dissipation $\overline{\epsilon}$. In the stationary state it reduces to a balance (146) between radial profiles that are each measured independently, so its residual tests the profiles; its dimensional form (140) is the starting point of §6.2. We proceed in three steps:

1. write the dimensional budget by adapting the dissipation equation of [3], (140);
2. nondimensionalize it into the self-similar frame, (143);
3. evaluate each term as a radial profile and form the stationary balance, (146).

Assumptions. Throughout, $1/Re = 0$, and the subgrid model is the self-similar constant eddy viscosity (97); the budget is not formed under the Smagorinsky model. As in §6.1, the ensemble-mean velocity vanishes, $\overline{\tilde{u}_i} = 0$, so $\tilde{u}_i$ is the fluctuating velocity. We work inside the sphere $r \le r_0$, where $W = 0$, so the sponge does not enter the budget. The quantity whose budget we form is the trace of the dissipation tensor of §5.7.2, which by (132) and (133) with $1/Re = 0$ is

$$\epsilon^* = 2\nu_{\mathrm{t}}^*\frac{\partial\tilde{u}_i^*}{\partial x_m^*}\frac{\partial\tilde{u}_i^*}{\partial x_m^*}, \qquad \epsilon = 2\nu_{\mathrm{t}}\frac{\partial\tilde{u}_i}{\partial x_m}\frac{\partial\tilde{u}_i}{\partial x_m}, \qquad \epsilon^* = \frac{U^3}{L}\,\epsilon. \tag{138}$$

This is twice the pseudo-dissipation of [3] (denoted ϵ there) with the eddy viscosity in place of ν.

Dimensional budget. Under the self-similar constant eddy viscosity, $\nu_\mathrm{t} = C_\nu \Delta^2$ with $\Delta = \Delta x$ fixed in the self-similar frame. Since $\Delta^* = L\,\Delta$, the dimensional eddy viscosity is

$$\nu_\mathrm{t}^* = C_\nu \Delta^2\, UL = UL\,\nu_\mathrm{t}. \tag{139}$$

It is uniform in space, so the subgrid force is $\partial_j(2\nu_\mathrm{t}^* \tilde{S}_{ij}^*) = \nu_\mathrm{t}^*\,\partial_j\partial_j\tilde{u}_i^*$ and the LES momentum equation is the Navier–Stokes equation for the filtered field with ν replaced by ν_t^*. The one difference is that ν_t^* changes in time with UL, whereas the derivation of [3] takes ν constant. To account for this, write $\epsilon^* = \nu_\mathrm{t}^*\,G^*$ with $G^* = 2\,\partial_m\tilde{u}_i^*\,\partial_m\tilde{u}_i^*$, so that

$$\frac{\partial\overline{\epsilon^*}}{\partial t^*} = \frac{1}{\nu_\mathrm{t}^*}\frac{\mathrm{d}\nu_\mathrm{t}^*}{\mathrm{d}t^*}\,\overline{\epsilon^*} + \nu_\mathrm{t}^*\,\frac{\partial\overline{G^*}}{\partial t^*}.$$

The first term comes from the change of the eddy viscosity alone. In the second, $\partial G^*/\partial t^*$ involves the viscosity only as the instantaneous coefficient of the viscous term, so the derivation of [3] applies unchanged: it gives their dissipation equation for zero mean velocity, doubled because of the factor of two in (138), with ν replaced by ν_t^*. The dimensional budget is therefore

$$\frac{\partial\overline{\epsilon^*}}{\partial t^*} = \frac{1}{\nu_\mathrm{t}^*}\frac{\mathrm{d}\nu_\mathrm{t}^*}{\mathrm{d}t^*}\,\overline{\epsilon^*} + \tilde{P}_\epsilon^* + \tilde{T}_\epsilon^* + \tilde{\Pi}_\epsilon^* + \tilde{D}_\epsilon^* - \tilde{\Upsilon}^*, \tag{140}$$

with

$$\begin{aligned}
\tilde{P}_\epsilon^* &= -4\nu_\mathrm{t}^*\,\overline{\frac{\partial\tilde{u}_i^*}{\partial x_k^*}\frac{\partial\tilde{u}_i^*}{\partial x_m^*}\frac{\partial\tilde{u}_k^*}{\partial x_m^*}} && \text{turbulent production,}\\
\tilde{T}_\epsilon^* &= -\frac{\partial}{\partial x_k^*}\,\overline{\tilde{u}_k^*\epsilon^*} && \text{turbulent transport,}\\
\tilde{\Pi}_\epsilon^* &= -\frac{4\nu_\mathrm{t}^*}{\rho}\,\frac{\partial}{\partial x_k^*}\,\overline{\frac{\partial\tilde{p}^*}{\partial x_m^*}\frac{\partial\tilde{u}_k^*}{\partial x_m^*}} && \text{pressure transport,}\\
\tilde{D}_\epsilon^* &= \nu_\mathrm{t}^*\frac{\partial^2\overline{\epsilon^*}}{\partial x_k^*\partial x_k^*} && \text{viscous diffusion,}\\
\tilde{\Upsilon}^* &= 4\,(\nu_\mathrm{t}^*)^2\,\overline{\frac{\partial^2\tilde{u}_i^*}{\partial x_k^*\partial x_m^*}\frac{\partial^2\tilde{u}_i^*}{\partial x_k^*\partial x_m^*}} && \text{dissipation of } \epsilon^*.
\end{aligned} \tag{141}$$

The tilde marks terms computed from the filtered velocity and pressure with the eddy viscosity. $\tilde{P}_\epsilon^*$ is the turbulent production P_ϵ^4 of [3]; their mean-advection term and the productions P_ϵ^1, P_ϵ^2, and P_ϵ^3 involve the mean velocity and vanish. Compared with the equation of [3], the eddy-viscosity change is the only term the LES adds.

Nondimensionalization. We now move (140) into the self-similar frame, where the profiles are measured. The substitutions are $\tilde{u}_i^* = U\tilde{u}_i$, $x_j^* = Lx_j$, $\tilde{p}^* = \rho U^2\tilde{p}$, $\nu_\mathrm{t}^* = UL\,\nu_\mathrm{t}$ (139), and $\epsilon^* = (U^3/L)\,\epsilon$ (138).

- *Right-hand side.* Each of the five terms carries exactly the factor U^4/L^2: each velocity gradient contributes U/L, each further derivative $1/L$, ν_t^* contributes UL, and the pressure ρU^2. For example, $\tilde{P}_\epsilon^* = (UL)(U/L)^3\,\tilde{P}_\epsilon = (U^4/L^2)\,\tilde{P}_\epsilon$.

- *Eddy-viscosity change.* With $\mathrm{d}t^*/\mathrm{d}t = \tau$ and the rates (4),

$$\frac{1}{\nu_\mathrm{t}^*}\frac{\mathrm{d}\nu_\mathrm{t}^*}{\mathrm{d}t^*} = \frac{\sigma_U + \sigma_L}{\tau}, \tag{142}$$

  and with the clock $\tau = L/U$ (88) the term becomes $(U^4/L^2)(\sigma_U + \sigma_L)\,\overline{\epsilon}$.

- *Left-hand side.* The chain rule (6), with the amplitude U^3/L (rate $3\sigma_U - \sigma_L$) in place of U, gives

$$\frac{\partial\overline{\epsilon^*}}{\partial t^*} = \frac{U^4}{L^2}\left(\frac{\partial\overline{\epsilon}}{\partial t} + (3\sigma_U - \sigma_L)\,\overline{\epsilon} - \sigma_L\,x_j\frac{\partial\overline{\epsilon}}{\partial x_j}\right).$$

Dividing by U^4/L^2 and collecting the terms in $\bar\epsilon$, $(\sigma_U+\sigma_L)-(3\sigma_U-\sigma_L)=-2(\sigma_U-\sigma_L)$, gives

$$\frac{\partial\bar\epsilon}{\partial t}=\sigma_L\,x_j\frac{\partial\bar\epsilon}{\partial x_j}-2\,(\sigma_U-\sigma_L)\,\bar\epsilon+\tilde P_\epsilon+\tilde T_\epsilon+\tilde\Pi_\epsilon+\tilde D_\epsilon-\tilde\Upsilon, \tag{143}$$

with the nondimensional terms

$$\begin{aligned}
\tilde P_\epsilon&=-4\nu_{\mathrm t}\,\overline{\frac{\partial\tilde u_i}{\partial x_k}\frac{\partial\tilde u_i}{\partial x_m}\frac{\partial\tilde u_k}{\partial x_m}}, & \tilde T_\epsilon&=-\frac{\partial}{\partial x_k}\,\overline{\tilde u_k\epsilon}, & \tilde\Pi_\epsilon&=-4\nu_{\mathrm t}\,\frac{\partial}{\partial x_k}\,\overline{\frac{\partial\tilde p}{\partial x_m}\frac{\partial\tilde u_k}{\partial x_m}},\\
\tilde D_\epsilon&=\nu_{\mathrm t}\frac{\partial^2\bar\epsilon}{\partial x_k\partial x_k}, & \tilde\Upsilon&=4\nu_{\mathrm t}^2\,\overline{\frac{\partial^2\tilde u_i}{\partial x_k\partial x_m}\frac{\partial^2\tilde u_i}{\partial x_k\partial x_m}},
\end{aligned} \tag{144}$$

of the same form as (141) without stars. The two new terms on the right of (143) are the scaling terms of the budget. The first, $\sigma_L\,x_j\,\partial_j\bar\epsilon$, advects $\bar\epsilon$ radially as the frame dilates. The second combines the rescaling of ϵ with the eddy-viscosity change: since $\nu_{\mathrm t}$ is constant in the self-similar frame, ϵ scales as the square of the gradient scale U/L, which changes at the rate $\sigma_U-\sigma_L$.

Radial profiles. Each term of (143) is the ensemble average of a pointwise scalar field. Since differentiation commutes with the average, this holds for the divergence terms too, e.g. $\tilde T_\epsilon=\overline{-\partial(\tilde u_k\epsilon)/\partial x_k}$. Each term is therefore of the form $\overline{Q}$ of §5.7.1, with Q one of the integrands

$$\begin{aligned}
\epsilon&=2\nu_{\mathrm t}\,\frac{\partial\tilde u_i}{\partial x_m}\frac{\partial\tilde u_i}{\partial x_m}, & p_\epsilon&=-4\nu_{\mathrm t}\,\frac{\partial\tilde u_i}{\partial x_k}\frac{\partial\tilde u_i}{\partial x_m}\frac{\partial\tilde u_k}{\partial x_m}, & t_\epsilon&=-\frac{\partial(\tilde u_k\epsilon)}{\partial x_k},\\
\pi_\epsilon&=-4\nu_{\mathrm t}\,\frac{\partial\tilde u_i}{\partial x_m}\frac{\partial^2\tilde p}{\partial x_i\partial x_m}, & d_\epsilon&=\nu_{\mathrm t}\frac{\partial^2\epsilon}{\partial x_k\partial x_k}, & \upsilon&=4\nu_{\mathrm t}^2\,\frac{\partial^2\tilde u_i}{\partial x_k\partial x_m}\frac{\partial^2\tilde u_i}{\partial x_k\partial x_m}.
\end{aligned} \tag{145}$$

Here π_ϵ is the pressure term before it is written as a divergence; the two forms agree by (92). Each integrand is evaluated spectrally from a snapshot of the self-similar fields, with the pressure recovered as in Appendix B.3. By the isotropy assumption of §5.7.1 each term depends only on r. It is estimated by the spherical mean (125) followed by the time average (126), e.g. $\tilde P_\epsilon(r)\approx T^{-1}\int_0^T\langle p_\epsilon\rangle_r\,\mathrm dt$. The advection term is $x_j\,\partial\bar\epsilon/\partial x_j=r\,\mathrm d\bar\epsilon/\mathrm dr$; it is computed from the same spectral coefficients as $\langle\epsilon\rangle_r$ by (201), and vanishes at $r=0$. The two scaling terms are formed per sample, with that sample's step rates $\sigma_{X,k}$, and averaged afterwards, so the balance is checked without assuming that the rates are constant.

The stationary budget. In the stationary state $\partial\bar\epsilon/\partial t=0$ and the rates are constant, $\sigma_X=c_\tau\beta_X$ (§6.1.1), so by (143) the profiles satisfy

$$0=\sigma_L\,r\frac{\mathrm d\bar\epsilon}{\mathrm dr}-2\,(\sigma_U-\sigma_L)\,\bar\epsilon+\tilde P_\epsilon+\tilde T_\epsilon+\tilde\Pi_\epsilon+\tilde D_\epsilon-\tilde\Upsilon, \tag{146}$$

each term a function of r. Since every term is estimated independently, the residual of (146) measures the statistical and discretization error of the profiles.

Equation (146) is the budget of the continuous equations. The split scheme of §2 also changes ϵ inside $r\le r_0$ in two ways that it leaves out. The sponge forcing, after the 2/3 mask and the projection, rings into $r\le r_0$ although $W=0$ there. And the dilation of the scaling step pushes modes past the 2/3 cutoff, where the next physics step masks them away together with their ϵ. Both are artifacts of the discretization and should shrink as the resolution increases, but at $N=128$ they leave a residual of about a quarter of the largest term near the centre.

5.8 Constraint functionals

The length and velocity scales will be set such that the zeroth and normalized second radial moments of the resolved energy density $\tilde u_i\tilde u_i=2\tilde k$, twice the resolved kinetic energy per unit mass (135), are constant. Mathematically, these moments are defined as

$$F_1[\tilde u]=\int\tilde u_i\tilde u_i\,\mathrm d\mathbf x\equiv E_0,\qquad F_2[\tilde u]=\frac{1}{E_0}\int r^2\,\tilde u_i\tilde u_i\,\mathrm d\mathbf x\equiv E_2, \tag{147}$$

where $r^2 = x_j x_j$ is the squared distance from the centre of the domain (42), and the E_0 in F_2 is the target value (44), a constant. As in the 1D case, E_0 measures the amplitude of the field and E_2 its spread.

The functionals are built from the resolved field alone: the subgrid kinetic energy k_{sgs} of (136) is neglected, so E_0 is twice the resolved kinetic energy rather than that of the full field, and E_2 is the mean-square radius of the resolved energy. This is accurate to the extent that the resolved field carries most of the energy. We monitor the resolved fraction $K_{\mathrm{res}}/(K_{\mathrm{res}} + K_{\mathrm{sgs}})$, with $K_{\mathrm{res}} = \int \tilde{k}\,\mathrm{d}\mathbf{x}$ and $K_{\mathrm{sgs}} = \int k_{\mathrm{sgs}}\,\mathrm{d}\mathbf{x}$ from the estimate (137), a posteriori against the criterion that a well-resolved LES carries at least 80% of the kinetic energy in the resolved field [2].

Writing $E_0^{\mathrm{in}}, E_2^{\mathrm{in}}$ for the moments of the field being scaled and substituting $\mathbf{y} = \mathbf{x}\,\lambda_L$ in each integral, with $d = 3$, the scaled field carries

$$\int \frac{1}{\lambda_U^2}\,\tilde{u}_i\tilde{u}_i(\mathbf{x}\,\lambda_L)\,\mathrm{d}\mathbf{x} = \frac{1}{\lambda_U^2\lambda_L^3}\,E_0^{\mathrm{in}}, \tag{148}$$

$$\frac{1}{E_0}\int r^2\,\frac{1}{\lambda_U^2}\,\tilde{u}_i\tilde{u}_i(\mathbf{x}\,\lambda_L)\,\mathrm{d}\mathbf{x} = \frac{1}{\lambda_U^2\lambda_L^5}\,E_2^{\mathrm{in}}, \tag{149}$$

so the homogeneity exponents of (29) are

$$(p_1, q_1) = (2, 3), \qquad (p_2, q_2) = (2, 5), \tag{150}$$

with $p_1 q_2 - p_2 q_1 = 4 \neq 0$, so the constraint system (30) is uniquely solvable. Solving it,

$$\lambda_L = \sqrt{\frac{E_0\,E_2^{\mathrm{in}}}{E_0^{\mathrm{in}}\,E_2}}, \tag{151}$$

$$\lambda_U = \sqrt{\frac{E_0^{\mathrm{in}}}{E_0\,\lambda_L^3}}, \tag{152}$$

where the length ratio is identical in form to its 1D counterpart (76).

5.9 Initial condition

Because the curl of a vector feild is divergence-free we will define $\tilde{u}_i$ as the curl of some feild A_i

$$\tilde{u}_i = \varepsilon_{ijk}\frac{\partial A_k}{\partial x_j}, \qquad \text{equivalently} \qquad \hat{\tilde{u}}_i = \mathrm{i}\,\varepsilon_{ijk}\,\kappa_j\hat{A}_k. \tag{153}$$

$\tilde{u}_i$ is built on the grid (42) in four steps, followed by a rescaling.

1. *Noise.* Each component $A_k(\mathbf{x_n})$ is an independent standard normal, drawn from a seeded Mersenne Twister.

2. *Spectral filter.* The noise is given a spectrum,

$$\hat{A}_k \leftarrow \hat{A}_k\,e^{-\kappa^2/\kappa_p^2}. \tag{154}$$

3. *Window.* The potential is localized in physical space,

$$A_k \leftarrow A_k\,e^{-r^2/2\sigma^2}, \tag{155}$$

   with r the same distance from the centre of the domain that E_2 uses in (147).

4. *Rescaling.* Finally $\tilde{u}_i \leftarrow \tilde{u}_i\sqrt{E_0/F_1[\tilde{u}]}$, with F_1 evaluated as the Riemann sum (43), so that the field delivered to the integrator carries a prescribed E_0.

5.10 Spectral interpolation in three dimensions

The scaling operator (19) needs the same band-limited evaluation (27) as in one dimension, but at $d = 3$ a direct evaluation of it is no longer affordable, and its realization on the real transform differs enough from (78) to be worth writing out.

The sum on the real transform. Write $\hat{\tilde{u}}_i^{(k)}(\mathbf{j}) = N^{-3}\,\mathrm{RFFT}(\tilde{u}_k)_{\mathbf{j}}$ for the normalized coefficients, with the first axis carrying only the non-negative frequencies $j_1 = 0, 1, \ldots, N/2$ and the remaining two carrying the full signed set $j_d = 0, 1, \ldots, N/2, -(N/2-1), \ldots, -1$. Two corrections turn (27) into a sum over that layout. First, the omitted Hermitian conjugates are restored by the multiplicity

$$m_{j_1} = \begin{cases} 1, & j_1 = 0, \text{ or } j_1 = N/2 \text{ with } N \text{ even}, \\ 2, & \text{otherwise}, \end{cases} \tag{156}$$

exactly as the factor of 2 does in (78). Second, the transform phases its coefficients to the sample index n, which on the centred grid (42) places its origin at $x = -H/2$ rather than at $x = 0$. Since the scaling operator dilates about the centre of the domain, each mode must be evaluated at $\lambda_L x + H/2$, not at $\lambda_L(x + H/2)$ — the two agree only at $\lambda_L = 1$. As $H/2$ is half a period, the correction is the constant phase $\exp(\mathrm{i}\kappa_j H/2) = (-1)^j$ per axis. Together,

$$\Phi^{(\mathrm{S})}_{\Delta t}(\tilde{u}_k)_i(\mathbf{x}) = \frac{1}{\lambda_{U,k+1}}\,\mathrm{Re}\sum_{j_1=0}^{N/2}\sum_{j_2}\sum_{j_3} m_{j_1}\,\hat{\tilde{u}}_i^{(k)}(\mathbf{j})\prod_{d=1}^{3}(-1)^{j_d}\exp\!\left(\frac{2\pi\mathrm{i}\,j_d\,\lambda_{L,k+1}\,x_d}{H}\right). \tag{157}$$

Separable evaluation. Evaluated as written, (157) costs $O(N^{2d})$: every retained mode is summed at every grid point, which at $N = 128$ is 4×10^{12} complex multiply–adds per scaling substep. The kernel, however, is a product of one factor per axis, so the triple sum separates and can be applied one axis at a time. Define the two interpolation matrices

$$W^{(1)}_{nj_1} = m_{j_1}(-1)^{j_1}\exp\!\left(\frac{2\pi\mathrm{i}\,j_1\lambda_{L,k+1}x_n}{H}\right), \qquad W^{(\mathrm{f})}_{nj} = (-1)^j\exp\!\left(\frac{2\pi\mathrm{i}\,j\,\lambda_{L,k+1}x_n}{H}\right), \tag{158}$$

the first of size $N \times (N/2+1)$ for the half-spectrum axis and the second of size $N \times N$ for the two full axes, both rebuilt whenever $\lambda_{L,k+1}$ changes. Then (157) is the sequence of contractions

$$\hat{\tilde{u}}_i \xrightarrow{W^{(\mathrm{f})} \text{ on axis } 3} \xrightarrow{W^{(\mathrm{f})} \text{ on axis } 2} \xrightarrow{W^{(1)} \text{ on axis } 1} \mathrm{Re}(\cdot), \tag{159}$$

each of which is a matrix–matrix product once the contracted axis is brought to the front. The cost falls to $O(d\,N^{d+1})$, i.e. 8×10^8 multiply–adds at $N = 128$, a factor of 5×10^3 below the direct sum, and the work is expressed as three dense GEMM calls. The factorization is exact — it is a regrouping of the same finite sum, not an approximation — so the guarantee of §1.3 that the scaling step contributes no spatial discretization error is retained. For $d = 1$ the sequence (159) collapses to the single contraction with $W^{(1)}$, which is (78).

The Nyquist bin. For even N the bins $j_d = \pm N/2$ of a real field's transform coincide, and the two axis layouts resolve that differently. On the half-spectrum axis the mode is stored once and is given multiplicity $m_{N/2} = 1$ by (156). On a full axis it is stored in the single bin $j_d = N/2$ but represents the conjugate pair, so it must be split evenly between $+N/2$ and $-N/2$; the two halves combine to

$$W^{(\mathrm{f})}_{n,\,N/2} = (-1)^{N/2}\cos\!\left(\frac{\pi N\lambda_{L,k+1}x_n}{H}\right), \tag{160}$$

the same cosine that closes (78). With (160) in place the contraction (159) reproduces the direct sum (157) to roundoff for either parity of N.

The out-of-box cutoff. The box indicator χ of (28) is a product of one factor per axis, like the kernel, so it separates the same way. It is imposed by zeroing row n of each interpolation matrix (158) whenever the query of that axis leaves the box,

$$W^{(1)}_{nj_1} \leftarrow 0, \quad W^{(\mathrm{f})}_{nj} \leftarrow 0 \qquad \text{whenever } |x_n \lambda_{L,k+1}| > H/2. \tag{161}$$

After the three contractions (159) the point $\mathbf{x_n}$ then carries the product of the three row factors, which is $\chi(\mathbf{x_n}\lambda_{L,k+1})$ exactly. The cutoff is thus applied at no cost beyond building the matrices, and the direct sum it is tested against applies χ pointwise. For $\lambda_{L,k+1}$ within a few percent of unity the zeroed rows are the outermost one or two grid points per side. These lie deep in the sponge layer (§5.3), where the field has already been damped, and ApplyBC, when enabled, clamps most of them anyway (§5.11). Zeroing a region in physical space is not a solenoidal operation, so, like the clamp, it reintroduces a divergence proportional to the amplitude it removes.

Loss of solenoidality and the Leray projection. The continuous scaling operator preserves continuity: if $\partial\tilde{u}_j/\partial x_j = 0$, then $\partial/\partial x_j\big[\lambda_U^{-1}\tilde{u}_j(\mathbf{x}\lambda_L)\big] = \lambda_U^{-1}\lambda_L\,(\partial\tilde{u}_j/\partial x_j)(\mathbf{x}\lambda_L) = 0$, the counterpart in physical space of (114). Its discrete evaluation does not preserve it exactly. The dilated field carries the wavenumbers $\lambda_L\boldsymbol{\kappa}$, which lie off the lattice of the grid, so sampling (157) at the grid points and transforming back redistributes each mode over neighbouring lattice wavenumbers without regard to $\kappa_i\hat{\tilde{u}}_i = 0$; the out-of-box cutoff (161) further removes amplitude in physical space. Both leave a divergence on the grid. When the option *project* of §2.5 is enabled, Scale therefore ends with the Leray projection of the scaled field back onto the solenoidal subspace, using the projection tensor (118),

$$\hat{\tilde{u}}_i(\boldsymbol{\kappa}) \leftarrow P_{il}(\boldsymbol{\kappa})\,\hat{\tilde{u}}_l(\boldsymbol{\kappa}) = \hat{\tilde{u}}_i - \frac{\kappa_i\kappa_l\hat{\tilde{u}}_l}{\kappa^2}, \qquad \boldsymbol{\kappa} \neq \mathbf{0}, \tag{162}$$

followed by the transform back to physical space. The mean mode $\boldsymbol{\kappa} = \mathbf{0}$ is left unchanged. The projection is orthogonal, so it can only remove energy, and the cost is one forward and one inverse transform per scaling step. Under Strang it is applied after each of the two scaling half steps, so the field entering Evolve is solenoidal as well as the field the step returns.

Interaction with dealiasing. The dilation maps each represented wavenumber to $\lambda_{L,k+1}\boldsymbol{\kappa}$, so a field that fills the spectrum up to the Nyquist wavenumber would wrap under any $\lambda_{L,k+1} > 1$. The 2/3 rule already required by the pseudo-spectral product (105) supplies the margin: the top retained mode sits at $N/3$ and moves to $\lambda_{L,k+1}N/3$, which stays below $N/2$ for every $\lambda_{L,k+1} < 3/2$. Runs in the self-similar frame hold $\lambda_{L,k+1}$ within a few percent of unity, so the margin is not approached, but it is the condition under which the interpolation stays free of aliasing. Finally, (159) is applied to each velocity component independently and mixes neither components nor the wavenumbers within an axis; at the exceptional value $\lambda_{L,k+1} = 1$ it reduces to the plain inverse transform, which is taken instead at $O(N^d \log N)$.

5.11 Boundary treatment

The clamp. When enabled, ApplyBC imposes a homogeneous Dirichlet condition on the velocity by clamping the outer layer of grid points of the centred domain (42) after each completed step,

$$\tilde{u}_i(\mathbf{x_n}) \leftarrow 0 \qquad \text{whenever } n_d \in \{0,\, N-1\} \text{ for some direction } d, \tag{163}$$

for every component $i = 1, 2, 3$. As in one dimension, on the centred grid $x_d \in [-H/2, H/2)$ the layers $n_d = 0$ and $n_d = N - 1$ are distinct grid points rather than periodic images of one another, so both are clamped; the six layers together are the boundary of the box.

Zeroing the faces is not a solenoidal operation, and ApplyBC does not project afterwards: the Leray projection (162), when enabled, is applied only inside Scale (§5.10). The clamp runs after the last scaling step of the step, so the stored field $\tilde{u}_{k+1}$ keeps its faces exactly zero and carries the small divergence the clamp injects there. Under Strang the first scaling half step of the next step projects it away before Evolve; under LieTrotter Evolve starts from it. The divergence stays small only while the field is negligible at the faces, which the sponge forcing (§5.3) is there to ensure.

6 Navier–Stokes results

6.1 Kinetic energy budget in the self-similar frame

We work with the full velocity u_i^* and the ensemble average $\overline{\cdot}$ of §5.7.1, and assume that the ensemble-mean velocity vanishes, $\overline{u_i^*} = 0$. Then u_i^* is the fluctuating velocity, the mean-advection and production terms of the turbulent kinetic energy equation [2] drop out, and the mean of the kinetic energy $\overline{k^*} = \frac{1}{2}\overline{u_i^* u_i^*}$, obeys

$$\frac{\partial \overline{k^*}}{\partial t^*} = \overline{T^*} - \frac{1}{2}\overline{\epsilon^*}, \qquad \overline{T^*} = -\frac{\partial}{\partial x_j^*}\left(\frac{1}{\rho}\overline{p^* u_j^*} + \frac{1}{2}\overline{u_i^* u_i^* u_j^*} - \nu \frac{\partial \overline{k^*}}{\partial x_j^*}\right), \tag{164}$$

where $\overline{T^*}$ is the transport and $\overline{\epsilon^*}$ is the trace of the dissipation tensor (128). The viscous term of the momentum equation contributes $\nu\, u_i^* \nabla^{*2} u_i^* = \nu \nabla^{*2} \overline{k^*} - \frac{1}{2}\overline{\epsilon^*}$: a viscous diffusion of $\overline{k^*}$, which we include in the transport, and the sink. Since $\overline{\epsilon^*}$ is the sink of $\overline{u_i^* u_i^*} = 2\overline{k^*}$, the sink of $\overline{k^*}$ is $\frac{1}{2}\overline{\epsilon^*}$.

We nondimensionalize with (2), the pressure scale (84), and

$$\overline{k^*} = U^2\, \overline{k}, \qquad \overline{\epsilon^*} = \frac{U^3}{L}\, \overline{\epsilon}, \tag{165}$$

as in (135) and (133). Since U and L are the same for every member of the ensemble (assumption 1 of §5.7.1), they pass through the averages, e.g. $\overline{u_i^* u_i^* u_j^*} = U^3\, \overline{u_i u_i u_j}$. The pressure and triple-correlation fluxes in $\overline{T^*}$ therefore carry U^3, the viscous flux carries $\nu U^2/L$, and each derivative contributes $\partial/\partial x_j^* = L^{-1}\, \partial/\partial x_j$, so that

$$\overline{T^*} = \frac{U^3}{L}\, \overline{T}, \qquad \overline{T} = -\frac{\partial}{\partial x_j}\left(\overline{p u_j} + \frac{1}{2}\overline{u_i u_i u_j} - \frac{1}{Re}\frac{\partial \overline{k}}{\partial x_j}\right), \tag{166}$$

with Re the Reynolds number (90). In the limit $1/Re = 0$ in which we run, the viscous term drops out of $\overline{T}$, while $\overline{\epsilon}$ stays finite and is estimated from the resolved field by (134).

The left-hand side of (164) follows from the chain rule exactly as in (6), with the amplitude U^2 in place of U, so that σ_U is replaced by $(U^2)'/U^2 = 2\sigma_U$,

$$\frac{\partial \overline{k^*}}{\partial t^*} = \frac{U^2}{\tau}\left(\frac{\partial \overline{k}}{\partial t} + 2\sigma_U\, \overline{k} - \sigma_L\, x_j \frac{\partial \overline{k}}{\partial x_j}\right). \tag{167}$$

With the clock (88), $U^2/\tau = U^3/L$, so the prefactors of (167) and (166) cancel and (164) becomes

$$\frac{\partial \overline{k}}{\partial t} = \sigma_L\, x_j \frac{\partial \overline{k}}{\partial x_j} - 2\sigma_U\, \overline{k} + \overline{T} - \tfrac{1}{2}\overline{\epsilon}. \tag{168}$$

6.1.1 Stationary budget

In the self-similar frame the flow is statistically stationary, so $\overline{k}$, $\overline{T}$, and $\overline{\epsilon}$ are independent of t and (168) reduces to

$$0 = \sigma_L\, x_j \frac{\partial \overline{k}}{\partial x_j} - 2\sigma_U\, \overline{k} + \overline{T} - \tfrac{1}{2}\overline{\epsilon}. \tag{169}$$

Since $\overline{T} - \frac{1}{2}\overline{\epsilon}$ is independent of t, so are the scaling terms. Under the power-law ansatz the rates are $\sigma_X = \beta_X \tau/(t^* + \alpha_X)$ by (52). Taking $\alpha_U = \alpha_L = \alpha$, the clock (88) is $\tau = c_\tau\, (t^* + \alpha)^{\beta_L - \beta_U}$ with

$$c_\tau = \frac{A_L}{A_U}, \tag{170}$$

and

$$\sigma_L\, x_j \frac{\partial \overline{k}}{\partial x_j} - 2\sigma_U\, \overline{k} = c_\tau\, (t^* + \alpha)^{\beta_L - \beta_U - 1}\left(\beta_L\, x_j \frac{\partial \overline{k}}{\partial x_j} - 2\beta_U\, \overline{k}\right). \tag{171}$$

Either the bracket vanishes identically, which by (169) would require $\overline{T} = \frac{1}{2}\overline{\epsilon}$ everywhere, or the power of $t^* + \alpha$ vanishes, $\beta_L - \beta_U = 1$; the numerical experiments confirm the latter. The clock is then $\tau = c_\tau(t^* + \alpha)$, the rates

$$\sigma_X = c_\tau\, \beta_X \tag{172}$$

are constant, the self-similar time is $t = c_\tau^{-1} \ln(t^* + \alpha)$ up to a constant, and (169) holds with constant coefficients,

$$0 = \sigma_L \, x_j \frac{\partial \overline{k}}{\partial x_j} - 2\sigma_U \, \overline{k} + \overline{T} - \tfrac{1}{2}\overline{\epsilon}. \tag{173}$$

The constant c_τ is not fixed by the analysis. The frame normalization $U_0 = L_0 = 1$ would give $c_\tau = 1$ only if the power law held from the initial condition on, which it does not, and the runs give c_τ well away from 1. Since $\sigma_L - \sigma_U = c_\tau(\beta_L - \beta_U) = c_\tau$, the mean rate difference over the fitted range and the ratio A_L/A_U of the fitted prefactors (§3.2) are two estimates of the same constant, a consistency check on the fit alongside $\beta_L - \beta_U = 1$ and $\alpha_U = \alpha_L$.

In physical variables, $\overline{T^*} - \frac{1}{2}\overline{\epsilon^*} = (U^3/L)(\overline{T} - \frac{1}{2}\overline{\epsilon})$ by (165) and (166), and $U^3/L = U^2/\tau = (A_U^2/c_\tau)\,(t^* + \alpha)^{2\beta_U - 1}$, so (164) and (173) give

$$\frac{\partial \overline{k^*}}{\partial t^*} = \frac{A_U^2}{c_\tau} \, (t^* + \alpha)^{2\beta_U - 1} \left(2\sigma_U \, \overline{k} - \sigma_L \, x_j \frac{\partial \overline{k}}{\partial x_j} \right). \tag{174}$$

At the origin $x_j \, \partial\overline{k}/\partial x_j = 0$, and (173) gives the energy there directly,

$$\overline{k}(0) = \frac{\overline{T}(0) - \frac{1}{2}\overline{\epsilon}(0)}{2\sigma_U}. \tag{175}$$

Then (174), with $\sigma_U = c_\tau \beta_U$, integrates to $\overline{k^*}(0, t^*) = A_U^2 \, (t^* + \alpha)^{2\beta_U} \, \overline{k}(0)$, the scaling (124) at $r = 0$ with $m_U = 2$, $m_L = 0$.

Away from the origin we use the isotropy assumption of §5.7.1, so that $\overline{k}$, $\overline{T}$, and $\overline{\epsilon}$ depend only on $r = \sqrt{x_i x_i}$. Then $x_j \, \partial\overline{k}/\partial x_j = r \, \mathrm{d}\overline{k}/\mathrm{d}r$, and (173) becomes the ordinary differential equation

$$\sigma_L \, r \frac{\mathrm{d}\overline{k}}{\mathrm{d}r} - 2\sigma_U \, \overline{k} = \tfrac{1}{2}\overline{\epsilon} - \overline{T}. \tag{176}$$

The ratio of the rates is that of the exponents, $\sigma_U/\sigma_L = \beta_U/\beta_L$ by (172), and with the integrating factor $r^{-2\beta_U/\beta_L}$ the equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}r} \left(r^{-2\beta_U/\beta_L} \, \overline{k} \right) = \frac{1}{\sigma_L} \, r^{-2\beta_U/\beta_L - 1} \left(\tfrac{1}{2}\overline{\epsilon} - \overline{T} \right). \tag{177}$$

The homogeneous solution is $\overline{k} = C \, r^{2\beta_U/\beta_L}$. For a flow whose length scale grows and whose velocity scale decays, $\beta_L > 0 > \beta_U$, and this is singular at the origin. Requiring $\overline{k}$ to be finite there sets $C = 0$, and integrating (177) from 0 to r gives

$$\overline{k}(r) = \frac{1}{\sigma_L} \int_0^r \left(\frac{s}{r} \right)^{-2\beta_U/\beta_L} \left(\tfrac{1}{2}\overline{\epsilon}(s) - \overline{T}(s) \right) \frac{\mathrm{d}s}{s}. \tag{178}$$

Finally, with $\overline{k^*} = U^2 \, \overline{k}$ and $r = r^*/L$, the physical energy is

$$\overline{k^*}(r^*, t^*) = A_U^2 \, (t^* + \alpha)^{2\beta_U} \, \overline{k}\left(\frac{r^*}{A_L \, (t^* + \alpha)^{\beta_L}} \right). \tag{179}$$

6.2 Dissipation budget

We compare the LES budget (140) of §5.7.3 with the budget of a DNS, under the same assumptions: the self-similar constant eddy viscosity, $1/Re = 0$, zero ensemble-mean velocity, and $r \le r_0$. Under the power-law ansatz (31), the eddy viscosity (139) has the logarithmic rate

$$\frac{1}{\nu_\mathrm{t}^*} \frac{\mathrm{d}\nu_\mathrm{t}^*}{\mathrm{d}t^*} = \frac{1}{U} \frac{\mathrm{d}U}{\mathrm{d}t^*} + \frac{1}{L} \frac{\mathrm{d}L}{\mathrm{d}t^*} = \frac{\beta_U}{t^* + \alpha_U} + \frac{\beta_L}{t^* + \alpha_L}, \tag{180}$$

and taking $\alpha_U = \alpha_L = \alpha$ as in §6.1.1, (140) becomes

$$\frac{\partial \overline{\epsilon^*}}{\partial t^*} = \frac{\beta_U + \beta_L}{t^* + \alpha} \, \overline{\epsilon^*} + \tilde{P}_\epsilon^* + \tilde{T}_\epsilon^* + \tilde{\Pi}_\epsilon^* + \tilde{D}_\epsilon^* - \tilde{\Upsilon}^*. \tag{181}$$

The equivalent terms of a DNS follow from the derivation of (140) with the full velocity u_i^*, the full pressure p^*, and the molecular viscosity ν, with ϵ^* replaced by the integrand of (128), $\epsilon^*_{\mathrm{DNS}} = 2\nu\,(\partial u_i^*/\partial x_m^*)(\partial u_i^*/\partial x_m^*)$; they are the terms of [3], again with the factor of two. The differences between the two sets are

$$
\begin{aligned}
\tilde{P}^*_\epsilon - P^*_\epsilon &= -4\left(\nu_{\mathrm{t}}^*\,\overline{\frac{\partial\tilde{u}_i^*}{\partial x_k^*}\frac{\partial\tilde{u}_i^*}{\partial x_m^*}\frac{\partial\tilde{u}_k^*}{\partial x_m^*}} - \nu\,\overline{\frac{\partial u_i^*}{\partial x_k^*}\frac{\partial u_i^*}{\partial x_m^*}\frac{\partial u_k^*}{\partial x_m^*}}\right),\\
\tilde{T}^*_\epsilon - T^*_\epsilon &= -\frac{\partial}{\partial x_k^*}\left(\overline{\tilde{u}_k^*\epsilon^*} - \overline{u_k^*\epsilon^*_{\mathrm{DNS}}}\right),\\
\tilde{\Pi}^*_\epsilon - \Pi^*_\epsilon &= -\frac{4}{\rho}\frac{\partial}{\partial x_k^*}\left(\nu_{\mathrm{t}}^*\,\overline{\frac{\partial\tilde{p}^*}{\partial x_m^*}\frac{\partial\tilde{u}_k^*}{\partial x_m^*}} - \nu\,\overline{\frac{\partial p^*}{\partial x_m^*}\frac{\partial u_k^*}{\partial x_m^*}}\right),\\
\tilde{D}^*_\epsilon - D^*_\epsilon &= \frac{\partial^2}{\partial x_k^* \partial x_k^*}\left(\nu_{\mathrm{t}}^*\,\overline{\epsilon^*} - \nu\,\overline{\epsilon^*_{\mathrm{DNS}}}\right),\\
\tilde{\Upsilon}^* - \Upsilon^* &= 4\left((\nu_{\mathrm{t}}^*)^2\,\overline{\frac{\partial^2\tilde{u}_i^*}{\partial x_k^*\partial x_m^*}\frac{\partial^2\tilde{u}_i^*}{\partial x_k^*\partial x_m^*}} - \nu^2\,\overline{\frac{\partial^2 u_i^*}{\partial x_k^*\partial x_m^*}\frac{\partial^2 u_i^*}{\partial x_k^*\partial x_m^*}}\right).
\end{aligned}
\tag{182}
$$

Each difference has two sources: ν_{t}^* in place of ν, and the filtered fields in place of the full fields. The closure (130) makes $\overline{\epsilon^*}$ an estimate of $\overline{\epsilon^*_{\mathrm{DNS}}}$, but it does not constrain the higher-order correlations, so the differences (182) need not vanish.

6.2.1 Residuals with respect to DNS

We compare the LES budget (181) with the budget of a DNS. The same derivation with the full velocity u_i^*, the full pressure p^*, and the constant molecular viscosity ν gives the budget of $\overline{\epsilon^*_{\mathrm{DNS}}}$, in which the eddy-viscosity change is absent,

$$
\frac{\partial\overline{\epsilon^*_{\mathrm{DNS}}}}{\partial t^*} = P^*_\epsilon + T^*_\epsilon + \Pi^*_\epsilon + D^*_\epsilon - \Upsilon^*,
\tag{183}
$$

with the terms of [3] entering (182). Subtracting (183) from (181),

$$
\frac{\partial}{\partial t^*}\left(\overline{\epsilon^*} - \overline{\epsilon^*_{\mathrm{DNS}}}\right) = \frac{\beta_U+\beta_L}{t^*+\alpha}\,\overline{\epsilon^*} + \left(\tilde{P}^*_\epsilon - P^*_\epsilon\right) + \left(\tilde{T}^*_\epsilon - T^*_\epsilon\right) + \left(\tilde{\Pi}^*_\epsilon - \Pi^*_\epsilon\right) + \left(\tilde{D}^*_\epsilon - D^*_\epsilon\right) - \left(\tilde{\Upsilon}^* - \Upsilon^*\right).
\tag{184}
$$

We now assume that the LES estimate is exact, $\overline{\epsilon^*} = \overline{\epsilon^*_{\mathrm{DNS}}}$, at every point and time inside $r \le r_0$. The left-hand side of (184) then vanishes, and the differences (182) are tied to the eddy-viscosity change,

$$
\left(\tilde{P}^*_\epsilon - P^*_\epsilon\right) + \left(\tilde{T}^*_\epsilon - T^*_\epsilon\right) + \left(\tilde{\Pi}^*_\epsilon - \Pi^*_\epsilon\right) + \left(\tilde{D}^*_\epsilon - D^*_\epsilon\right) - \left(\tilde{\Upsilon}^* - \Upsilon^*\right) = -\frac{\beta_U+\beta_L}{t^*+\alpha}\,\overline{\epsilon^*}.
\tag{185}
$$

For the LES to reproduce $\overline{\epsilon^*_{\mathrm{DNS}}}$, the residuals of its budget terms must therefore cancel the eddy-viscosity change, the one term of (181) without a DNS counterpart.

A The RK4 physics step

This appendix sets out the scheme used for the Navier–Stokes physics operator $\Phi^{(\mathrm{P})}$ of §5.6: one step of the classical fourth-order Runge–Kutta method. Throughout we abbreviate the step as $h \equiv \Delta t$, reserving Δ for the filter width of the subgrid stress model of §5.2.

A.1 The scheme

Written out mode by mode, the physics flow (120) is the system of ordinary differential equations

$$
\frac{\mathrm{d}\hat{\tilde{u}}_i}{\mathrm{d}t} = R_i(\hat{\tilde{u}}), \qquad R_i(\hat{\tilde{u}}) = -P_{il}(\boldsymbol{\kappa})\left(\hat{N}_l - \hat{G}_l + \hat{F}_l\right),
\tag{186}
$$

the bracketed physical terms of (119). RK4 advances it through four evaluations of R,

$$\begin{aligned} \hat{k}_i^{(1)} &= R_i\big(\hat{\tilde{u}}^{(n)}\big), & \hat{k}_i^{(2)} &= R_i\big(\hat{\tilde{u}}^{(n)} + \tfrac{h}{2}\hat{k}^{(1)}\big), \\ \hat{k}_i^{(3)} &= R_i\big(\hat{\tilde{u}}^{(n)} + \tfrac{h}{2}\hat{k}^{(2)}\big), & \hat{k}_i^{(4)} &= R_i\big(\hat{\tilde{u}}^{(n)} + h\,\hat{k}^{(3)}\big), \end{aligned} \tag{187}$$

combined with Simpson weights,

$$\hat{\tilde{u}}_i^{(n+1)} = \hat{\tilde{u}}_i^{(n)} + \frac{h}{6}\left(\hat{k}_i^{(1)} + 2\hat{k}_i^{(2)} + 2\hat{k}_i^{(3)} + \hat{k}_i^{(4)}\right). \tag{188}$$

The convective part of each evaluation is computed pseudo-spectrally in rotational form, $\tilde{u}_j\partial_j\tilde{u}_i = -(\tilde{\mathbf{u}} \times \tilde{\boldsymbol{\omega}})_i + \partial_i(\tilde{u}_k\tilde{u}_k/2)$, whose gradient part P_{il} annihilates; this costs six inverse and three forward transforms. Under the Smagorinsky model the subgrid stress term (108) costs six further inverse transforms, one per independent component of $\tilde{S}_{ij}$, and six forward transforms of T_{ij}; under the self-similar constant eddy viscosity it is the diagonal multiplication (109) and costs none. The sponge term (110) reuses the physical-space velocity already formed for the convective term and multiplies it by the tabulated W, so it costs only the three forward transforms of $W\tilde{u}_i$. An evaluation thus costs 24 transforms under the Smagorinsky model and 12 under the constant eddy viscosity, and a step 96 or 48 in all. The scheme has no coefficients beyond the four weights, so nothing is precomputed or cached.

Order. The scheme is fourth order in h, so it satisfies the second-order requirement §2.5 places on EVOLVE for the Strang composition with a margin.

A.2 Stability

Applied to the linear model $\mathrm{d}\hat{\tilde{u}}/\mathrm{d}t = \lambda\hat{\tilde{u}}$, (187)–(188) multiply $\hat{\tilde{u}}$ by the amplification factor

$$A(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}, \qquad z = \lambda h, \tag{189}$$

the degree-four Taylor polynomial of e^z, and the step is stable while $|A(z)| \le 1$.

The advective limit. The convective term is, locally, a transport at speed $|\tilde{\mathbf{u}}|$, whose eigenvalues on the retained band lie on the imaginary axis with $|\lambda| \le \max|\tilde{\mathbf{u}}|\,\kappa_{\max}$, where $\kappa_{\max} = 2\pi(N/3)/H$ is the top mode retained by the 2/3 rule. Along the imaginary axis $|A(z)| \le 1$ holds for $|z| \le 2\sqrt{2}$, so

$$h\,\max|\tilde{\mathbf{u}}|\,\kappa_{\max} \le 2\sqrt{2}. \tag{190}$$

The self-similar frame pins $\max|\tilde{\mathbf{u}}| = O(1)$ on average, so the limit does not drift over a run as it would in a decaying simulation in physical variables. It does not pin it pointwise in time, however: the field can momentarily concentrate into a high-amplitude structure whose $\max|\tilde{\mathbf{u}}|$ carries a step that is stable in the settled regime past (190).

The sponge limit. The sponge forcing (100) is, pointwise, a linear decay at rate $W(r) \in [0, W_{\max}]$, whose eigenvalues lie on the negative real axis with $|\lambda| \le W_{\max}$. Along the negative real axis $|A(z)| \le 1$ holds for $-2.785 \lesssim z \le 0$, so

$$h\,W_{\max} \le 2.785. \tag{191}$$

Unlike (190), this limit does not depend on the state: it is fixed by the run parameter $W_{\max}$. It is not enforced, but assumed. At $W_{\max} = 10$ and a step of $h \approx 0.05$, $h\,W_{\max} \approx 0.5$, well inside both (191) and the range $h\,W_{\max} \lesssim 1$ in which the step also follows the exponential decay e^{-Wh} accurately. A run with a larger $W_{\max}$ or a longer step must check (191) by hand.

The diffusive limit. Under the self-similar constant eddy viscosity, the subgrid stress term (109) is marched explicitly with the rest of R_i. It is a linear decay at rate $C_\nu \Delta^2 \kappa^2$, whose eigenvalues lie on the negative real axis with $|\lambda| \le C_\nu \Delta^2 \kappa_{\max}^2$, so by the same bound as (191)

$$h\, C_\nu \Delta^2 \kappa_{\max}^2 \le 2.785. \tag{192}$$

With $\Delta = \Delta x = H/N$ and $\kappa_{\max} = 2\pi(N/3)/H$, the product $\Delta\, \kappa_{\max} = 2\pi/3$ is independent of the grid, and the limit becomes

$$h\, C_\nu \le \frac{9 \times 2.785}{4\pi^2} \approx 0.635. \tag{193}$$

Like the sponge limit it does not depend on the state, since in the self-similar frame $\nu_{\mathrm{t}} = C_\nu \Delta^2$ is fixed for the run, and because the eddy viscosity scales with Δ^2 it does not tighten under grid refinement either. It is not enforced, but assumed, and holds for $\Delta t \le 0.635/C_\nu$. At a step of $h \approx 0.05$ it holds for $C_\nu \lesssim 12.7$.

The CFL number. The advective limit is therefore the only one that depends on the state. The step at which (190) holds with equality,

$$\Delta t_{\mathrm{CFL}}(\tilde{u}) = \frac{2\sqrt{2}}{\max |\tilde{\mathbf{u}}|\, \kappa_{\max}}, \tag{194}$$

is the bound of §2.3, and the realized CFL (45) is $\Delta t \max |\tilde{\mathbf{u}}_{k+1}|\, \kappa_{\max}/2\sqrt{2}$. At a constant step a momentary concentration is not stepped around; it shows up as a realized CFL above one.

A resolution margin. A structure that concentrates below the grid scale also raises $\max |\tilde{\mathbf{u}}|$, and so looks like a CFL problem; but a smaller step only delays it. To tell the two apart the run also records the fraction of the dealiased energy $\frac{1}{2}\sum |\hat{\tilde{u}}|^2$ carried by $|\boldsymbol{\kappa}| > \frac{2}{3}\kappa_{\max}$. It stays small while the field is resolved and rises as energy piles up against the 2/3 cutoff. A realized CFL excursion with a small tail fraction is a time-step problem; one preceded by a rising tail fraction is a resolution problem.

A.3 Solenoidality

The step contributes nothing of its own to the continuity residual. Every contribution to R_i — convective, subgrid, and sponge alike — has had P_{il} applied, and $\kappa_i P_{il} = \kappa_l - \kappa_l \kappa^2/\kappa^2 = 0$, so contracting (186) with κ_i gives $\kappa_i R_i(\hat{\tilde{u}}) = 0$. Each stage of (187) is then divergence free, and the divergence of each mode is carried through the step unchanged,

$$\kappa_i \hat{\tilde{u}}_i^{(n+1)} = \kappa_i \hat{\tilde{u}}_i^{(n)}. \tag{195}$$

Whatever divergence enters the field is therefore neither amplified nor damped by the step, and if (112) holds on input it holds on output. The residual the run log records is therefore entirely the boundary clamp's, when the clamp is enabled (§5.11), and not the time stepper's.

A.4 Dealiasing

The 2/3 mask of §5.4 is applied to the incoming field before the first stage. No stage then leaves the retained band: the convective, subgrid stress, and sponge terms are masked after their forward transforms, the projection P_{il} is diagonal in $\boldsymbol{\kappa}$, and each stage argument in (187) is a linear combination of band-limited fields. The mask is applied once more to $\hat{\tilde{u}}^{(n+1)}$, which changes nothing in exact arithmetic but makes explicit that no content reaches the band the next scaling step will stretch.

B Radial profiles from the spectral fields

This appendix sets out how the radial profiles of §5.7 are computed from a snapshot of the self-similar fields: the spherical mean (125) at a prescribed set of radii, the pressure that enters $\tilde{\Pi}_\epsilon$, the refined grid on which the integrands are formed, and the sampling in time.

B.1 The spherical mean of a band-limited field

Let $Q(\mathbf{x_n})$ be a real field on the grid (42), with discrete Fourier coefficients

$$\hat{Q}_{\mathbf{m}} = \sum_{\mathbf{n}} Q(\mathbf{x_n})\, e^{-\mathrm{i}\boldsymbol{\kappa}_{\mathbf{m}}\cdot\mathbf{n}\,\Delta x}, \qquad \boldsymbol{\kappa}_{\mathbf{m}} = \frac{2\pi}{H}\,\mathbf{m}, \tag{196}$$

over the signed indices $m_j \in \{-N/2, \ldots, N/2-1\}$. The transform is phased to the sample index, whose origin sits at $\mathbf{x} = -(H/2)(1,1,1)$ rather than at $\mathbf{x} = 0$, so its trigonometric interpolant is

$$Q(\mathbf{x}) = \frac{1}{N^3}\sum_{\mathbf{m}} \hat{s}_{\mathbf{m}}\, e^{\mathrm{i}\boldsymbol{\kappa}_{\mathbf{m}}\cdot\mathbf{x}}, \qquad \hat{s}_{\mathbf{m}} = (-1)^{m_1+m_2+m_3}\,\hat{Q}_{\mathbf{m}}, \tag{197}$$

with the phase-origin correction of (78) in each direction, since $\kappa_{m_j}H/2 = \pi m_j$. The spherical mean of a single mode is

$$\frac{1}{4\pi}\oint e^{\mathrm{i}\boldsymbol{\kappa}\cdot r\hat{\mathbf{n}}}\,\mathrm{d}\Omega = j_0(|\boldsymbol{\kappa}|\,r), \qquad j_0(z) = \frac{\sin z}{z}, \quad j_0(0) = 1, \tag{198}$$

so the spherical mean (125) of the interpolant is

$$\langle Q\rangle_r = \frac{1}{N^3}\sum_{\mathbf{m}} \mathrm{Re}\big(\hat{s}_{\mathbf{m}}\big)\, j_0(|\boldsymbol{\kappa}_{\mathbf{m}}|\,r), \tag{199}$$

the real part because Q is real and j_0 is even. Evaluated with the real transform, which stores the half spectrum $m_1 \geq 0$, each stored mode with $0 < m_1 < N/2$ stands also for its omitted conjugate and carries weight 2, as in (78); the planes $m_1 = 0$ and $m_1 = N/2$ carry weight 1. At $r = 0$, (199) is the interpolant at the origin, which is the grid value there.

Equation (199) is exact for the interpolant; no interpolation is done in physical space. For the resolved velocity, which is band-limited to the retained 2/3 band, the interpolant is the field itself. For a product such as ϵ, formed pointwise on the grid, it is the interpolant of the product as sampled, the same spectral interpolation as §1.3.

The kernel depends on $\boldsymbol{\kappa}_{\mathbf{m}}$ only through the integer $s = m_1^2+m_2^2+m_3^2 \leq 3(N/2)^2$. Summing the weighted coefficients over each such value first,

$$g_s = \sum_{\mathbf{m}:|\mathbf{m}|^2=s} c_{\mathbf{m}}\,\mathrm{Re}\big(\hat{s}_{\mathbf{m}}\big), \qquad \langle Q\rangle_r = \frac{1}{N^3}\sum_s g_s\, j_0\!\left(\frac{2\pi}{H}\sqrt{s}\,r\right), \tag{200}$$

with $c_{\mathbf{m}} \in \{1, 2\}$ the weights above, turns the sum into a regrouping of the spectrum, done once per field, followed by a product with the matrix $J_{is} = j_0\big((2\pi/H)\sqrt{s}\,r_i\big)$, tabulated once per run for the radii r_i below. The cost per field is one forward transform and a matrix–vector product of size $n_r \times 3(N/2)^2$.

The radial derivative of the profile follows from the same coefficients. Since $j_0' = -j_1$, with $j_1(z) = \sin z/z^2 - \cos z/z$,

$$r\frac{\partial\langle Q\rangle_r}{\partial r} = -\frac{1}{N^3}\sum_s g_s\,\kappa_s r\, j_1(\kappa_s r), \qquad \kappa_s = \frac{2\pi}{H}\sqrt{s}, \tag{201}$$

a second tabulated kernel applied to the same g_s. Because the spherical mean commutes with $x_j\,\partial/\partial x_j = r\,\partial/\partial r$, this is $\langle x_j\,\partial Q/\partial x_j\rangle_r$, the advection term of (143). It is evaluated this way rather than from the pointwise product $x_j\,\partial Q/\partial x_j$, which is not periodic and whose interpolant rings at the faces of the box. It vanishes at $r = 0$.

B.2 The radii

The profiles are evaluated at $r = 0$ and at n_r logarithmically spaced radii,

$$r_i = r_{\min}\left(\frac{r_{\max}}{r_{\min}}\right)^{(i-1)/(n_r-1)}, \qquad i = 1, \ldots, n_r, \tag{202}$$

together with the few radii at which the statistical error is reported, which are added to the set exactly rather than taken from the nearest r_i. By default $n_r = 32$, $r_{\min} = \Delta x$, and $r_{\max} = r_0$, the radius inside which the sponge forcing vanishes and the ϵ budget of §5.7.3 applies.

B.3 The pressure

The pressure transport $\tilde{\Pi}_\epsilon$ needs the nondimensional pressure itself, which the projection (118) eliminates and the solver never forms. It is recovered from the Poisson equation (116) with the convective term in divergence form, $\hat{N}_l = \mathrm{i}\kappa_j \widehat{\tilde{u}_l \tilde{u}_j}$, which equals (105) for a solenoidal field,

$$\hat{\tilde{p}} = -\frac{\kappa_l \kappa_j \, \widehat{\tilde{u}_l \tilde{u}_j}}{\kappa^2}, \qquad \kappa \neq 0, \qquad \hat{\tilde{p}}(\mathbf{0}) = 0, \tag{203}$$

without the sponge term $\mathrm{i}\kappa_l \hat{F}_l/\kappa^2$ of (116), since the sponge is not part of the budget (§5.7.3). The products $\tilde{u}_l \tilde{u}_j$ are formed on the refined grid of §B.4, where they are exact. The solver itself forms the convective term in rotational form, $-\tilde{\mathbf{u}} \times \boldsymbol{\omega}$, which differs from $\tilde{u}_j \, \partial \tilde{u}_i / \partial x_j$ by the gradient $\partial_i(\frac{1}{2}\tilde{u}_k \tilde{u}_k)$; the pressure it implies is therefore the total head $\tilde{p} + \frac{1}{2}\tilde{u}_k \tilde{u}_k$, not the $\tilde{p}$ of the convective form in which the budget (143) is written. The subgrid term $\mathrm{i}\kappa_l \hat{G}_l/\kappa^2$ is zero under the self-similar constant eddy viscosity, since $\kappa_l \hat{G}_l = 0$. The scaling terms contribute nothing to the pressure (§5.5).

B.4 The integrands

The integrands (145) are cubic in $\tilde{u}_i$, and on the solver's grid they alias badly. The cubic mask leaves the field filled up to $|m_d| = N/3$ in every direction, so a product of n such fields reaches $|m_d| = nN/3$, and a cubic one reaches the box wavenumber N, twice the Nyquist wavenumber $N/2$. The aliased content folds onto the represented modes and corrupts both the interpolant between grid points, which the spherical mean (200) uses, and every spectral derivative of a product. The divergence terms t_ϵ and π_ϵ suffer most.

We therefore form every product on a refined grid of $N_p = 2\lceil aN/2 \rceil$ points per direction, with padding factor $a \geq 1$, onto which the field is carried by zero-padding its spectrum. The normalized coefficients $\hat{Q}_{\mathbf{m}}/N^3$ of (196) carry over unchanged, so the field on the refined grid is the spectral interpolant of the stored one; the Nyquist modes of the solver's grid, which the mask keeps empty, are dropped. A product of n fields is then exact while $nN/3 < N_p/2$, that is for $a > 2n/3$: $a = 2$ makes every cubic integrand exact, and is the default. The eddy viscosity keeps the filter width of the run, $\Delta = H/N$, whatever N_p is, and $a = 1$ recovers evaluation on the solver's grid.

On the refined grid every derivative is taken spectrally, $\widehat{\partial Q/\partial x_j} = \mathrm{i}\kappa_j \hat{Q}$, and every product is formed pointwise in physical space. The divergence terms t_ϵ, π_ϵ, and d_ϵ are formed as products first and differentiated spectrally afterwards, so that each is the divergence of the grid flux and not a sum of separately interpolated terms. Each integrand is then reduced to its profile by (200) on the refined grid, which is exact there as on the solver's grid.

B.5 Sampling in time

The profiles cost of order a hundred transforms per snapshot on the refined grid, against the handful of a physics step on the solver's grid, so they are not evaluated every step. They are evaluated on the stored field u_k every n_s steps, $n_s = \mathrm{round}(\Delta t_s/\Delta t)$, so that the samples are equally spaced in t at the interval Δt_s. The interval need only be short against the integral time scale of the profiles, which the batch means of Appendix C measure; by default $\Delta t_s = 0.02$, an order of magnitude below the integral time scale of $\epsilon(0)$ observed in the runs.

C Statistical error of time averages

This appendix sets out how the statistical error of a time-averaged quantity such as (126) is estimated, following [4]. Let $q(t)$ be a statistically stationary signal, here the spherical mean (125) at some radius r of the trace of the nondimensional dissipation tensor, $q(t) = \langle \epsilon \rangle_r(t)$ of (134), or of the resolved kinetic energy, $q(t) = \langle \tilde{k} \rangle_r(t)$ of (135), both in the self-similar frame, with mean $\langle q \rangle$, fluctuation $q' = q - \langle q \rangle$, and root-mean-square fluctuation q'_{rms}. Because the clock (88) is the advective time scale L/U, the self-similar time t is already measured in eddy turnover times.

C.1 Batch means

Successive values of q are correlated, so the samples of a run cannot be treated as independent. We instead use the method of batch means:

1. Discard the initial transient, during which the flow approaches the self-similar state, and let T be the length of the remaining record.
2. Split the record into $M = T/T_w$ non-overlapping windows of length T_w, and compute the window means
$$\langle q\rangle_i = \frac{1}{T_w}\int_{t_i}^{t_i+T_w} q(t)\,\mathrm{d}t, \qquad i = 1,\dots,M. \tag{204}$$
The samples $q_k = q(t_k)$ are recorded at equally spaced t_k, once per step (§2.3) or once per fixed sampling interval. Each integral (204) is evaluated as the exact integral of the piecewise-linear interpolant of the samples: the trapezoid rule on (t_k, q_k), with q interpolated linearly to window edges that fall between samples. The windows then tile $[t_0, t_0 + MT_w]$ exactly, and the remainder of the record shorter than T_w is dropped.
3. Estimate the mean and its standard error (SEM) as
$$q_{\mathrm{est}} = \frac{1}{M}\sum_{i=1}^{M}\langle q\rangle_i, \qquad \mathrm{SEM} \simeq \frac{\mathrm{STD}(\langle q\rangle_i)}{\sqrt{M}}. \tag{205}$$
4. Report the 95% confidence interval $q_{\mathrm{est}} \pm 1.96\,\mathrm{SEM}$.

The confidence interval assumes a Gaussian error, which the central limit theorem supplies when the windows are uncorrelated. Equation (205) holds only if

1. the windows are uncorrelated, $T_w \gg \tau_{\mathrm{corr}}$, where τ_{corr} is the integral time scale of q; and
2. there are enough windows, $M \gg 1$.

For a fixed T the two conditions pull in opposite directions, so T_w has to be chosen with care.

C.2 Choosing the window length

The window length is chosen by plotting $\mathrm{SEM}\sqrt{T}$ against T_w, both in units of the eddy turnover time. The curve has three regimes:

- *Small T_w.* Adjacent windows are correlated, so M overcounts the independent samples and $\mathrm{SEM} \approx q'_{\mathrm{rms}}\sqrt{T_w/T}$, a slope of $1/2$ on log axes. This underestimates the true error.
- *Plateau.* Both conditions hold and $\mathrm{SEM} \approx q'_{\mathrm{rms}}\sqrt{\tau_{\mathrm{corr}}/T}$, independent of T_w. This is the true SEM.
- *Large T_w.* M is too small and the estimate is noisy and unreliable.

The small-T_w asymptote and the plateau intersect at $T_w = \tau_{\mathrm{corr}}$, where
$$\tau_{\mathrm{corr}} = \frac{1}{q'^2_{\mathrm{rms}}}\int_{-\infty}^{\infty} C(\tau)\,\mathrm{d}\tau, \qquad C(\tau) = \langle q'(t)\,q'(t+\tau)\rangle, \tag{206}$$
so the plot also yields the integral time scale. A clear plateau requires a long record: [4] found one only for $T \gtrsim 10^4$ eddy turnover times, approximately 1.6×10^4 for homogeneous isotropic turbulence and 1.4×10^4 for channel flow at $Re_\tau = 180$. A dissipation-type quantity, being a second-order statistic of the velocity gradients, has τ_{corr} of roughly five eddy turnover times in homogeneous isotropic turbulence.

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